

## Fouille de réseaux sociaux Social mining

C. Largeron  
Professeur d'Informatique  
Université Jean Monnet  
SE2 - LabHC  
04/77/91/57/56  
christine.Largeron@univ-st-etienne.fr

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## Who am I ? C. LARGERON

Professor of Computer Science, Université Jean Monnet  
Adjunct professor at University of Alberta, Canada

What am I doing:

Head of IEGI - Master IAE Institut

Research in Data mining

Text mining, social mining

Information retrieval

Hubert Curien Laboratory - UMR CNRS 5516

How to join me :

04/77/91/57/56

Christine.Largeron@univ-st-etienne.fr

WI - MLDM - M1

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## What are we doing together ?

Lessons 1 & 2: text mining (12 et 14 sept 2023- 9h-12h – L125)

Lessons 3 & 4: social mining (14 Nov 9h00 – 12h00, 23 Nov 9h-12h – L216)

Lessons 5&6: Practical lessons: R. Vaudaine

**Evaluation:** Exercices & examination & project

Examination: exercices based on the course (C.Largeron + B Jeudy)

Project: see later

WI - MLDM - M1

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## Outline

- Introduction
- Networks representation, storage and visualization
- Network analysis : measures and metrics
- Models of network generation
- Node Classification
- Community detection
- Influence/ link prediction
- Conclusion
- References
  
- Application with R

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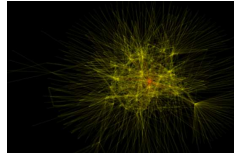
## Introduction : definition

### Definition (Wasserman & Faust,1994)

« **Social network: finite set or sets of actors and the relation or relations defined on them** »

Actors : entities, individuals, organizations, ..

Relations: friendship, kinship, ...



### Typical social networks:

- Social science: collaboration, work, economic exchange...
- biology : neural network, genes, proteines,
- linguistic : synonymie, co-occurrence, etc (cf. Wordnet)
- transport : road, air route,

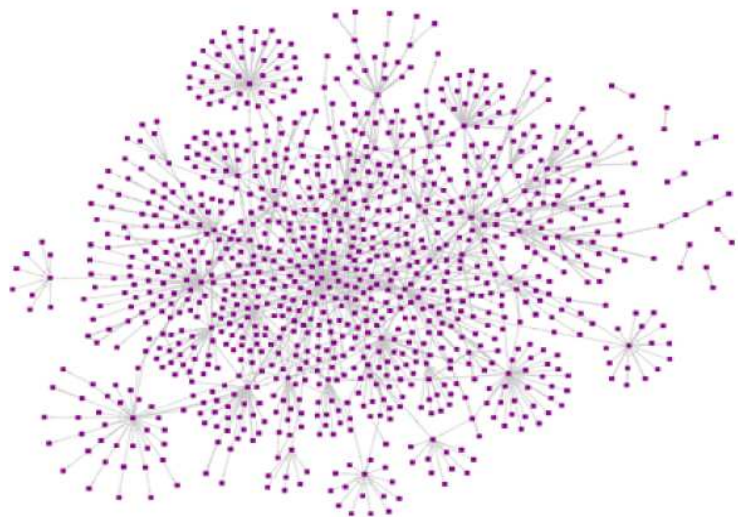
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## Introduction : social mining

- Blogs
  - Blogspot
  - Wordpress
- Wikis
  - Wikipedia
  - Wikiversity
- Social Networking Sites
  - Facebook
  - Myspace
  - Orkut
- Digital media sharing websites
  - Youtube
  - Flickr
- Social Tagging
  - Del.icio.us
- Others
  - Twitter
  - Yelp



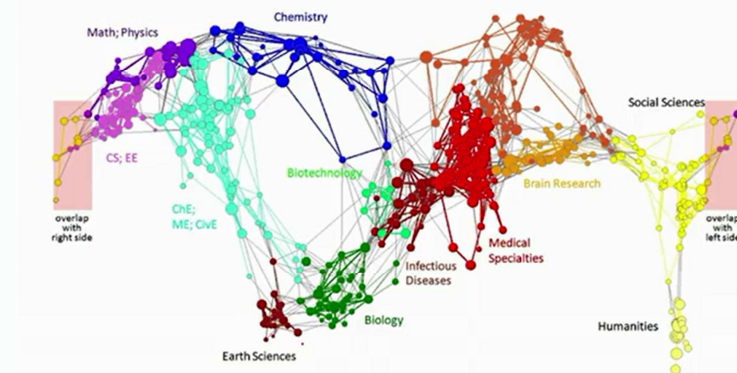
## Introduction: social mining



Relationship between key recovery agencies after Katrina  
© thinkNola.com and orgnet.com

## Introduction: social mining

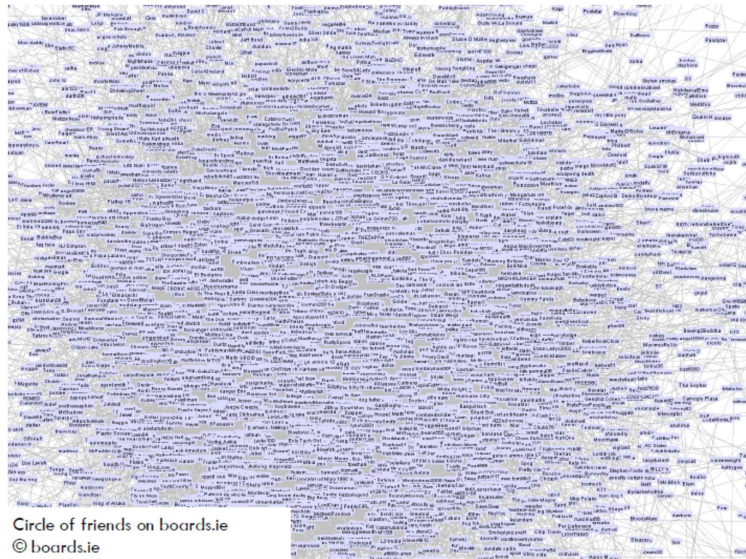
### Graph Data: Information Nets



Citation networks and Maps of science  
[Börner et al., 2012]

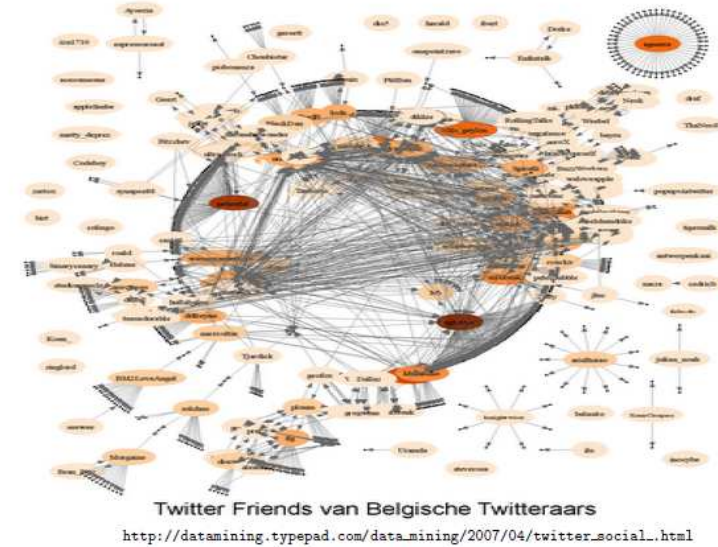
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## Introduction : What happens



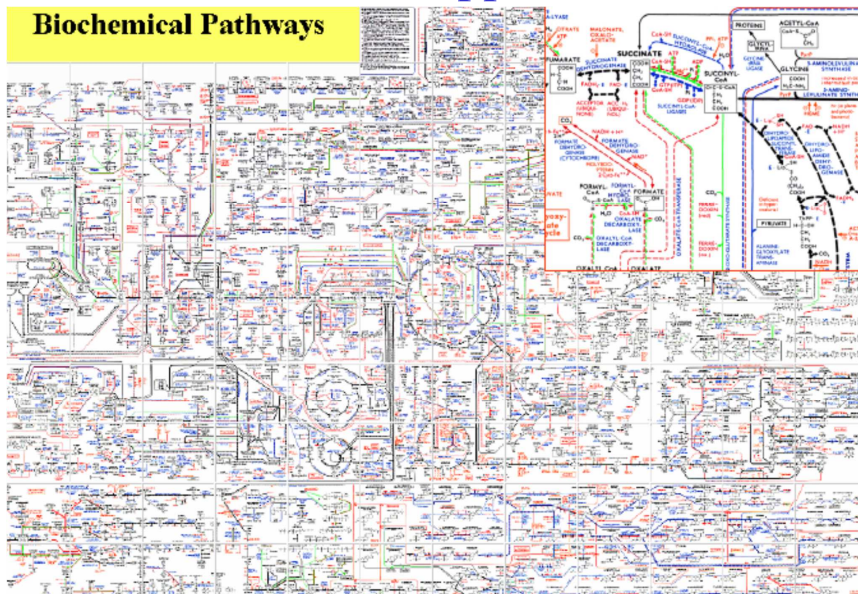
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## Introduction : What happens



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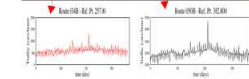
## Introduction : What happens



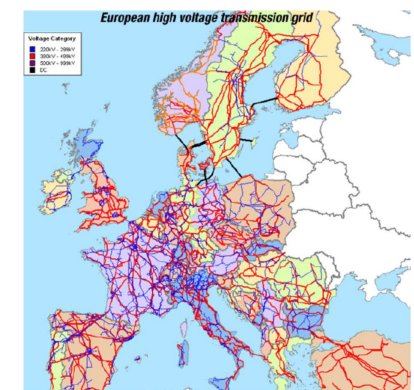
## Introduction : What happens

### Highways

$f_i(t)$  = traffic at a given point of a road  $i$  at day  $t$ .



Daily traffic on 127 Colorado roads from 1998 to 2001.



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## Introduction : from social network analysis to social mining

From an historical perspective

### Sociologic analysis

- Moreno 1933 (sociogram)
- Milgram 1967 (6 degrees of separation - small world)
- Cartwright and Harary, 1977 (graph theory)

### Web 2.0: Facebook, Myspace, etc

Renewed interest from the Web based on social networks websites development.

- Social bookmarking (Del.icio.us)
- Friendship networks (Facebook, Myspace, LinkedIn)
- Blogosphere
- Media Sharing (Flickr, Youtube)
- Folksonomies

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## Introduction : social network - > information network

### • Information network [Sun et al. 2009]

A network where each node represents an entity (e.g., actor in a social network) and each link (e.g., tie) a relationship between entities

- Each node/link may have attributes, labels, and weights
- Node / link may carry rich semantic information

### • Homogeneous vs. heterogeneous networks

- Homogeneous networks
  - Single object type and single link type
  - Single model social networks (e.g., friends)
  - WWW: a collection of linked Web pages
- Heterogeneous, multi-typed networks
  - Multiple object and link types
    - Medical network: patients, doctors, disease, contacts, treatments
    - Bibliographic network: publications, authors, venues

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## Introduction : What happens

### Social networks characteristics

- Low density,
- power law degree distribution,
- low diameter (six degree), ..

### Information networks characteristics

- Size
- Complex graphs (heterogeneous networks, attributed network, multiplex network etc.)

Roberto Interdonato, Martin Atzmueller, Sabrina Gaito, Rushed Kanawati, Christine Largeron, Alessandra Sala: Feature-rich networks: going beyond complex network topologies. Appl. Netw. Sci. 4(1): 4:1-4:13 (2019)

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## Introduction : What happens

### Social networks represent a financial market

- 14,27 Billions (\$) in 2014 – 70,7 in 2019 – 115 in 2021 : advertising gains for Facebook (@statistica)

### Emergence of new applications

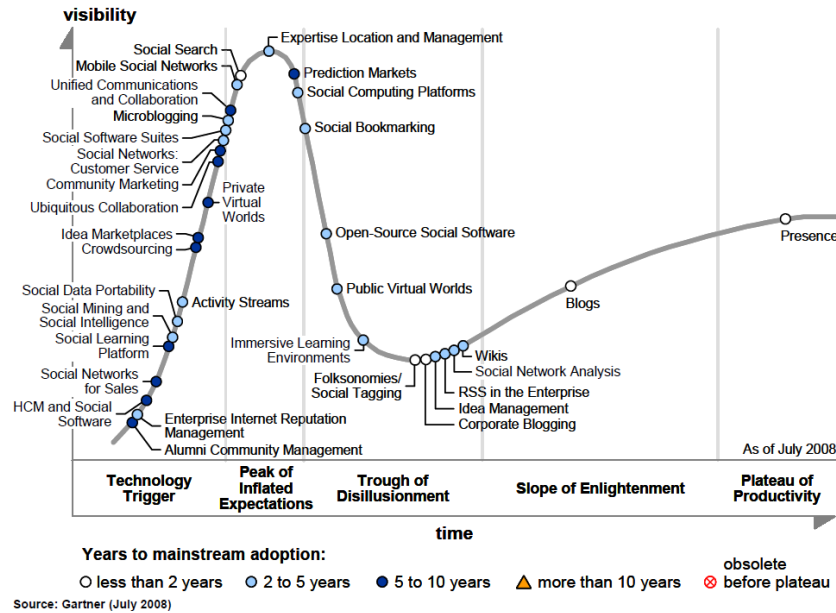
- “**By 2014, social networking services will replace e-mail as the primary vehicle for interpersonal communications for 20 percent of business users.**” (Gartner 2008)
- Social network analysis is getting **mature**.

These applications need adapted software.

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# Introduction : What happens

Figure 1. Hype Cycle for Social Software, 2008



# Top 10 Data and Analytics Technology Trends That Will Change Your Business Published: 11 April 2019

## Table Of Contents

### Analysis

- What Is Driving the Top 10 Data and Analytics Technology Trends?
- Trend 1: Augmented Analytics
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- Trend 4: Graph Analytics
- Trend 5: Commercial AI/ML Will Dominate the Market Over Open Source
- Trend 6: Data Fabric
- Trend 7: Explainable AI
- Trend 8: Blockchain in Data and Analytics
- Trend 9: Continuous Intelligence
- Trend 10: Persistent Memory Servers

<https://www.gartner.com/en/documents/3906812/top-10-data-and-analytics-technology-trends-that-will-ch>

# Introduction : What happens



# Introduction : What happens

## Wikipedia : Social network analysis software

Wikipedia: Social network analysis software

From Wikipedia, the free encyclopedia

This article may require cleanup to meet Wikipedia's quality standards. Please improve this article if you can. The talk page may contain suggestions. *(June 2008)*

Social network tools are:

- For scholarly research tools like UCInet<sup>[1]</sup>, Pajek<sup>[2]</sup>, ORA<sup>[3]</sup>, the statnet<sup>[4]</sup> suite of packages in R, and GUESS<sup>[5]</sup> are p
- Examples of business oriented social network tools include iPoint<sup>[6]</sup>, NetMiner<sup>[7]</sup>, InFlow<sup>[8]</sup>, KeyHub<sup>[9]</sup>, Sentinel Visual KXEN Social Network<sup>[10]</sup>, NodeXL<sup>[11]</sup>. For large networks with millions of nodes, try Sonoma<sup>[12]</sup> or ORA<sup>[13]</sup>. For mobile telephones Iridio SNA Plus<sup>[14]</sup> is recommended.
- An open source package with GUI for Linux, Windows and Mac, is Social Networks Visualizer or SocNetV<sup>[15]</sup>, developed QIC++.
- Another generic open source package for Windows, Linux and OS X with interfaces to Python and R is "igraph"<sup>[16]</sup>
- Another generic open source package with (GUI) for Windows, Linux and OS X is "Tulip"
- Rapinet<sup>[17]</sup> is a generic freely available open source solution for network analysis and interactive visual network exploration.
- For Mac OS X a related package installer of SocNetV<sup>[18]</sup> is available.
- For integrated egocentric data collection and visualization<sup>[19]</sup>

Gephi, igraph with R , NetworkX, Tulip, etc.

## Introduction : What happens? Need registration yet !

**International network for social network analysis**  
<http://www.insna.org/software/index.html>



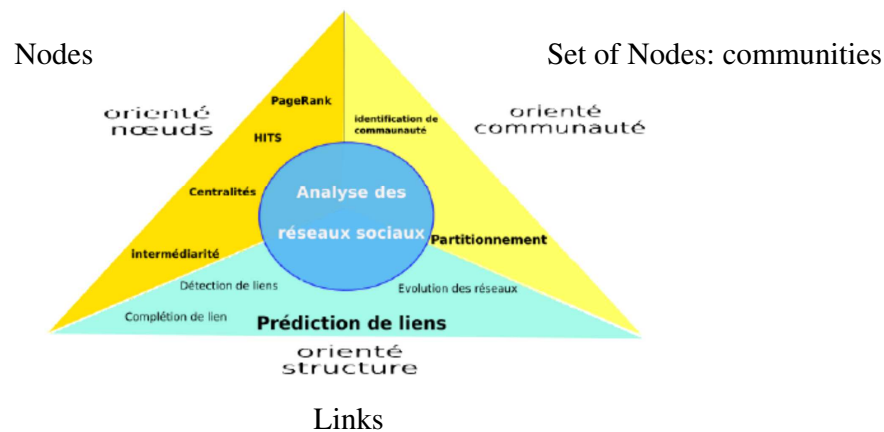
## Introduction : Social mining,why ?

Understand / predict the comportment of the entities

- Representation / visualization of the network
  - Graph theory, ...
- Description and analysis of the network at the global level or at the local level (Social network analysis )
  - Measures and metrics
- Simulation of networks
  - Network generation models
- Detection of group of entities with strong relations
  - Community detection
- Dynamic analysis
  - Node classification, link prediction, social influence analysis, diffusion analysis, ...

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# Introduction



@ Rushed Kanawati

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# Outline

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  - Influence/ link prediction
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- 
- Application with R

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## Networks representation as graph (Cartwright and

Harary, 1977)

Social network: finite set of actors and the relation defined on them

### Graph representation

$G = (V, E)$  with

$V$  : set of nodes or vertices with  $n = |V|$

$E \subseteq (V \times V)$  : set of links with  $m = |E|$

where

nodes = actors = entities

links (edges or arcs) = relationships

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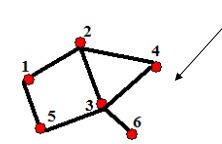
## Networks representation as graph

(Cartwright and Harary, 1977)

- Multi-edge:
  - if more than one edge between the same pair of vertices
  - Useful to analyse more than one relation
- Self-edge (self-loop): if an edge connects vertex to itself
- Simple network: if a network has neither self-edges nor multi-edges
- Example

$G = (E, V)$  with  $E = \{1, 2, 3, 4, 5, 6\}$

and  $V = \{(1, 2), (1, 5), (2, 3), (2, 4), (3, 4), (3, 5), (3, 6)\}$



$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

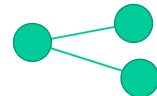
Adjacency matrix

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## Networks representation

- Link orientation
  - Undirected links : **edges** (co-authorship)

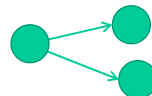
$$\forall (x, y) \in E, (y, x) \in E$$



In the next: undirected graph !

- Directed link : each **arc** has a direction (e-mails sent)

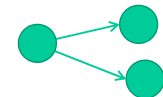
$$(x, y) \neq (y, x)$$



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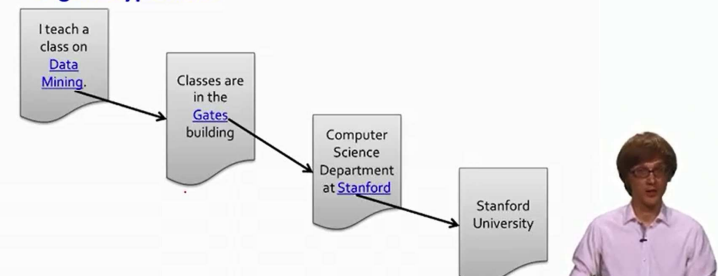
## Networks representation

- Link orientation: Directed link
  - $(x, y) \neq (y, x)$



### Web as a directed graph:

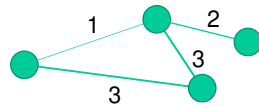
- Nodes: Webpages
- Edges: Hyperlinks



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## Networks representation

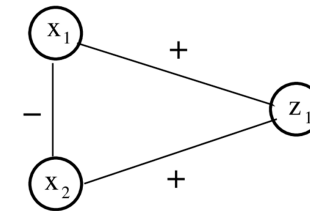
- **Weighted graph:** links having weight, usually a real number that represents the strength of the relation between two nodes
- $G = (V, E, L)$  with
  - $V$  : set of nodes (or
  - $E \subseteq (V \times V)$  : set of links
  - $L : E \rightarrow \mathbb{R}$
  - $u=(x,y) \rightarrow L(u)$



Weights and thickness of the line

## Networks representation

- **General case:** weight takes positive or null values  
 $L(u) \in \mathbb{R}^+$
- **Specific case:** signed graphs [Harary, 1954]
  - weight takes positive or negative values or signs
  - used to model opposite relationships (friend/enemy)



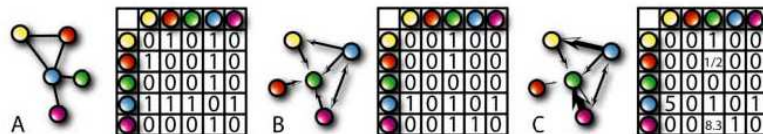
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## Networks representation as graph and adjacency matrix

- Undirected
- Simple graph
- Symmetric

Directed

Directed  
Weighted

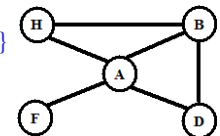


Type of graphs and the adjacency matrix associated (square matrix)

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## Networks representation as graph

- Let a network  $G = (V, E)$
- **Undirected Network (considered in the sequel)**
  - Degree (or degree centrality) of a vertex:  $d(v_i) \in \{0, n-1\}$   
 $d(v_i) = |v_j| \text{ s.t. } e_{ij} \in E \wedge e_{ij} = e_{ji}$
  - # of edges connected to it, e.g.,  $d(A) = 4$ ,  $d(H) = 2$



- Directed network

- In-degree of a vertex  $d_{in}(v_i)$ :

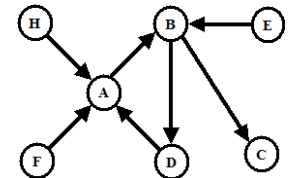
$$d_{in}(v_i) = |v_j| \text{ s.t. } e_{ji} \in E$$

- # of edges pointing to  $v_i$  E.g.,  $d_{in}(A) = 3$ ,  $d_{in}(B) = 2$

- Out-degree of a vertex  $d_{out}(v_i)$ :

$$d_{out}(v_i) = |v_j| \text{ s.t. } e_{ij} \in E$$

- # of edges from  $v_i$  E.g.,  $d_{out}(A) = 1$ ,  $d_{out}(B) = 2$



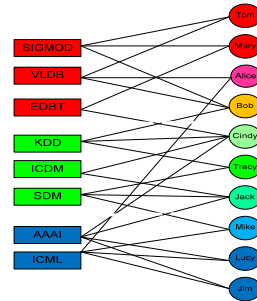
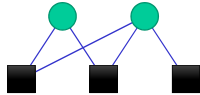
- **Neighboring nodes** : x and y are neighboring nodes if there is a link between x and y

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## Networks representation

- Bipartite network or two modes : two kinds of vertices, and edges linking only vertices of unlike types

- $G = (T, \perp, E \subseteq T \times \perp)$  with  $T \cap \perp = \emptyset$



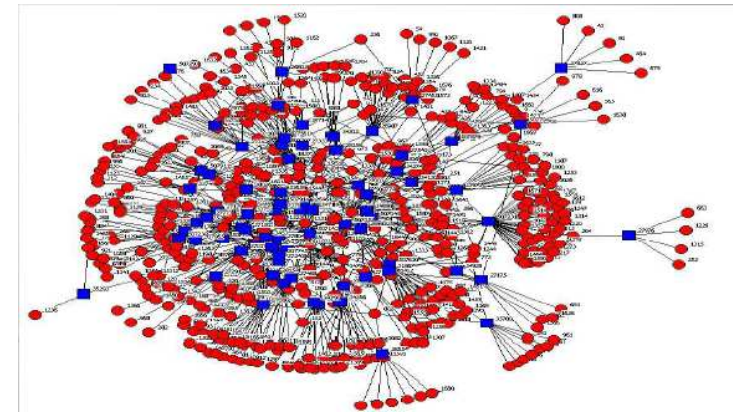
- Incidence matrix ( $\Leftrightarrow$  Square matrix):

$B_{ij} = 1$  if vertex  $i \in T$  links to  $j \in \perp$ ; 0 otherwise

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## Network representation

Bipartite or two modes graph: Collaborations in PCRD



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## Networks representation

- One can create a one-mode project from the two-mode partite form (but with information loss)

$$G_T^n = (V_T \subseteq T, E_T = \{x, y \in T : |\Gamma_G(x) \cap \Gamma_G(y)| \geq n\})$$

$$G_\perp^m = (V_\perp \subseteq \perp, E_\perp = \{x, y \in \perp : |\Gamma_G(x) \cap \Gamma_G(y)| \geq m\})$$

where  $n$  and  $m$  are parameters

- The projection to one-mode can be written in terms of the incidence matrix  $\mathbf{B}$  as follows

$$P_{ij} = \sum_{k=1}^g B_{ki} B_{kj} = \sum_{k=1}^g B_{ik}^T B_{kj}$$

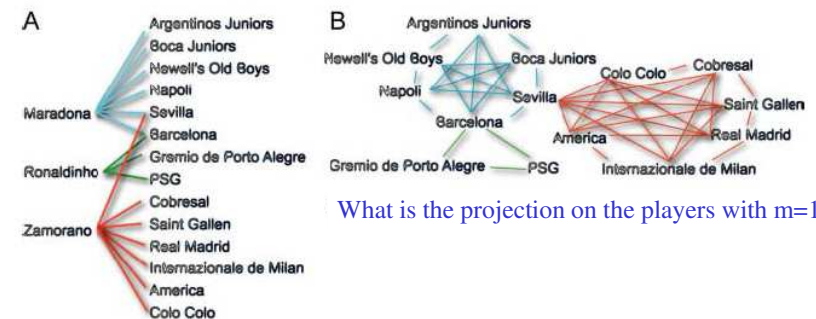
The product of  $B_{ki} B_{kj}$  will be 1 if  $i$  and  $j$  both belong to the sample group  $k$  in the bi-partite network

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## Network representation

A Bipartite and its projected graphs

What is the projection on the teams with  $n=1$  ?

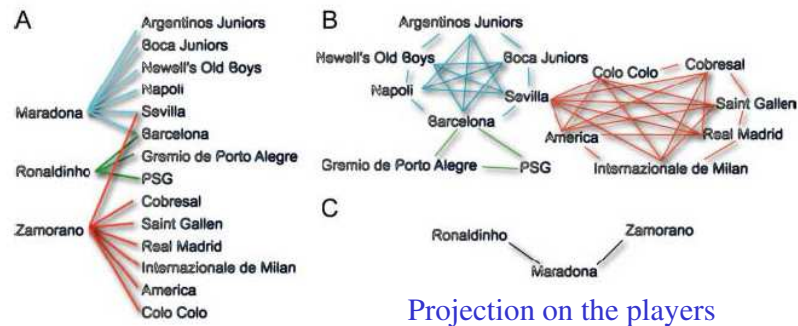


What is the projection on the players with  $m=1$ ?

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# Network representation

A Bipartite and its projected graphs

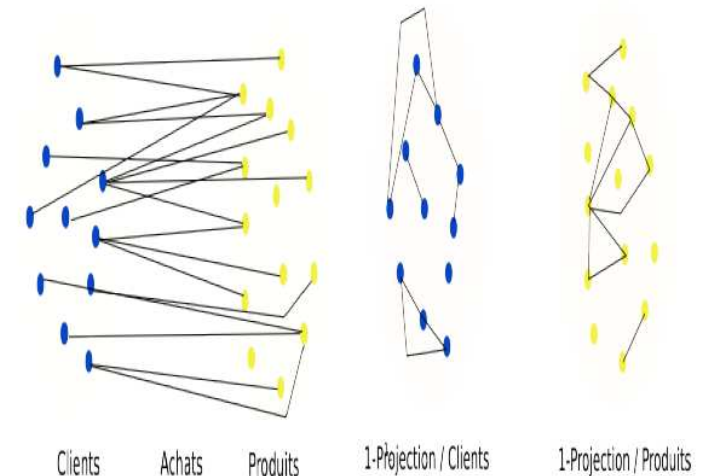


Projection on the players

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# Network representation

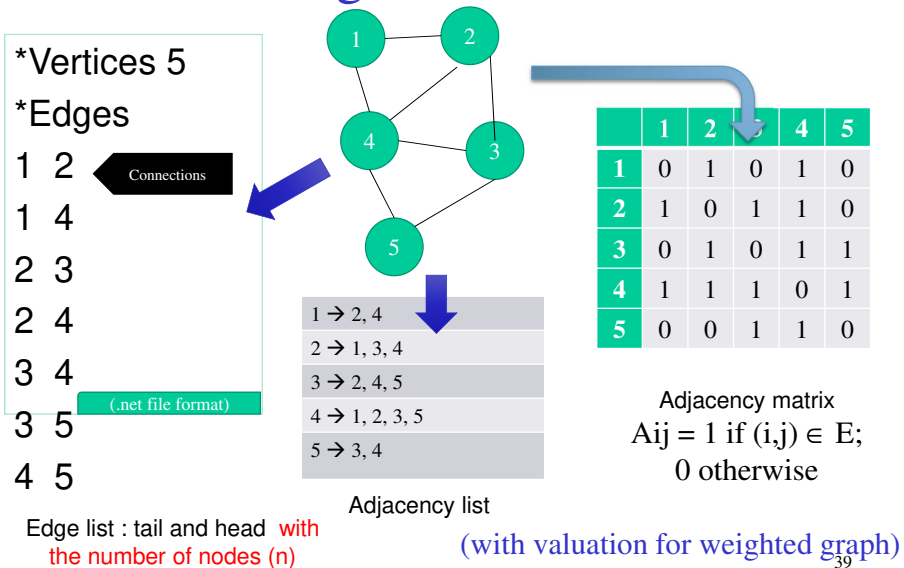
A Bipartite and its projected graphs for recommendation systems



Which application? Customers buy products

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# Network storage



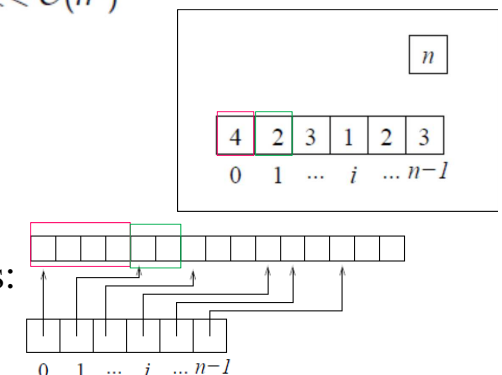
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# Network storage

Social networks are not dense: Sparse adjacency matrix

$$\rightarrow \mathcal{O}(m) \ll \mathcal{O}(n^2)$$

- Number of nodes
- Degree of each node:  
number of successors
- List of the neighbours:  
list of successors



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## Network storage

Usual input / output format (type of files)

- Pajek file format (.net)
  - Matrix, edge/arc list, weights
- GML ( Graph modelling Language)
  - Edge/arc list, node/edge label, nodes coordinates
- GraphML
  - Edge/arc list, hypergraph, attribute
- DL (Data language – UCINET - Pajek)
  - Matrix, edge/arc list, two-mode graph
- Gexf (Gephi)
- ...

All are based on the previous representations (matrix, list) + meta data

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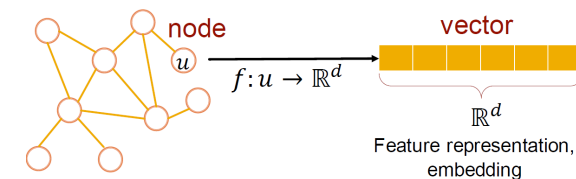
## Graph embedding

### Graph representation learning

**Aim** : Converting graphs or parts of a graph (nodes, edges, substructures) into low-dimensional, dense and continuous vector space while preserving structural properties.

**Graph embedding**: each graph (of a set) is encoded into a vector such that two graphs having the same structure are near together in the vectorial space

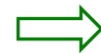
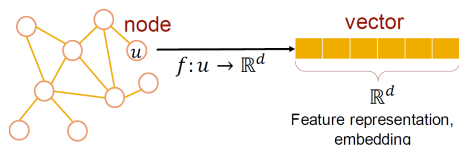
**Node embedding**: each node is encoded into a vector such that two nodes having **similar property** are near together in the vectorial space



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## Graph embedding

**Interest**: Allows to apply to graph data all the range of data mining and machine learning techniques designed for tabular data (vectors).



### Tasks

- Node classification
- Link prediction
- Graph classification
- Anomalous node detection
- Clustering
- ....

**How?** (Cui, TKDE2018) :

- Matrix factorization
- Random walk (Node2vec, Deepwalk)
- Deep learning methods

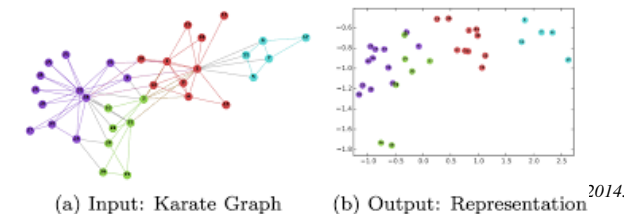
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## Node embedding

**Node embedding**: each graph is encoded into a vector such that two nodes having **similar property** are near together in the vectorial space

**Example of property**:

- first order proximity or neighborhood
- second order proximity or structural equivalence
- isomorphic equivalence
- etc.



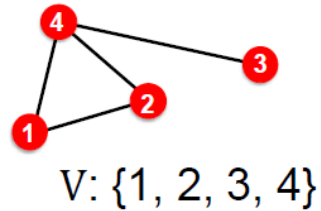
2014.

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## Node embedding

### Assume we have a graph G:

- V is the vertex set.
- A is the adjacency matrix (assume binary).
- For simplicity: No node features or extra information is used



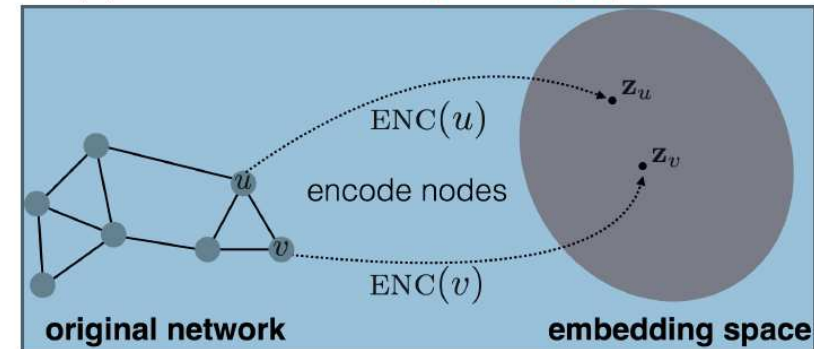
$$A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

@Leskovec,2021

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## Node embedding

- Goal is to encode nodes so that similarity in the embedding space (e.g., dot product) approximates similarity in the graph



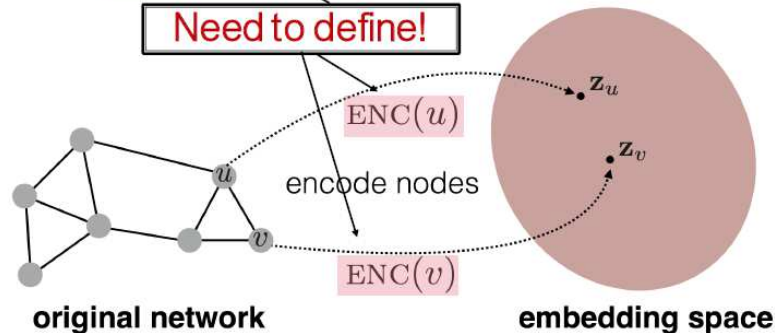
@Leskovec,2021

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## Node embedding

Goal:  $\text{similarity}(u, v)$  in the original network  $\approx \mathbf{z}_v^T \mathbf{z}_u$  Similarity of the embedding

Need to define!



@Leskovec,2021

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## Node embedding

1. **Encoder** maps from nodes to embeddings
2. **Define a node similarity function** (i.e., a measure of similarity in the original network)
3. **Decoder DEC** maps from embeddings to the similarity score
4. **Optimize the parameters of the encoder so that:**

$$\text{similarity}(u, v) \text{ in the original network} \approx \mathbf{z}_v^T \mathbf{z}_u \text{ Similarity of the embedding}$$

DEC( $\mathbf{z}_v^T \mathbf{z}_u$ )

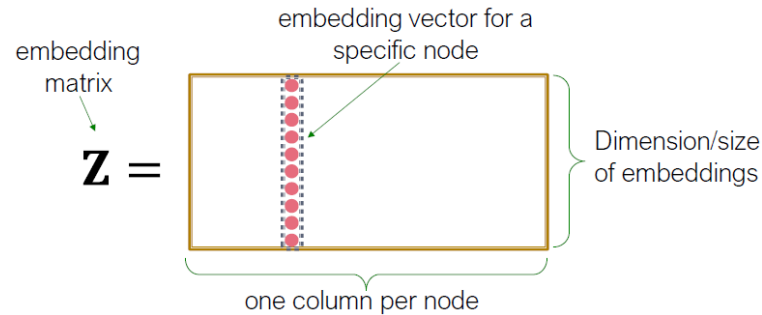
@Leskovec,2021

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# Node embedding

Methods based on random walks

- DeepWalk (Perozzi, 2014)
- Node2Vec (Grover, 2016)



@Leskovec,2021

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# Node embedding

| Name of the method                        | Graph sim.           | Embedding sim. |
|-------------------------------------------|----------------------|----------------|
| Laplacian Eigenmaps (LE) [1] - $O(N^2)$   | 1st-order prox       | Euclidean      |
| Locally Linear Emb. (LLE) [17] - $O(N^2)$ | 1st-order prox       | Euclidean      |
| HOPE [14] - $O(N^2)$                      | Katz-Index           | Dot-product    |
| SVD of the adjacency matrix - $O(N^2)$    | 2nd-order prox       | Dot-product    |
| struc2vec (S2V) [15] - $O(N \log(N))$     | Co-occurrence proba  | Dot-product    |
| node2vec (N2V) [9] - $O(N)$               | Co-occurrence proba  | Dot-product    |
| Verse [18] - $O(N)$                       | Perso. Page-Rank     | Dot-product    |
| Kamada-Kawai layout (KKL) [11] - $O(N^2)$ |                      | Euclidean      |
| Multi-dim Scaling (MDS) [12]              | 1st-order prox       | Euclidean      |
| SDNE [19] - $O(N)$                        | 1st & 2nd-order prox | Euclidean      |

@Vaudaine, IDA2019

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## Graph embedding surveys

References:

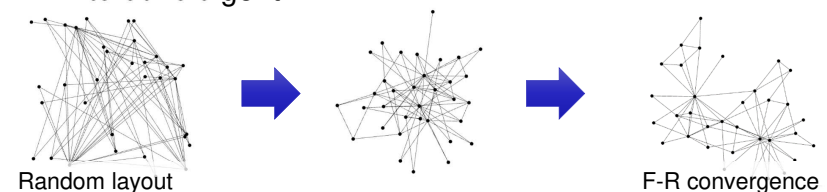
- Cai, Z., Chang, K.: A comprehensive survey of graph embedding: problems, techniques, and applications. TKDE 30(9), 1616–1637 (2018)
- Cui, P., Wang, X., Pei, J., Zhu, W.: A survey on network embedding. CoRR, abs/1711.08752 (2017)
- Goyal, P., Ferrara, E.: Graph embedding techniques, applications, and performance: a survey. Knowl.-Based Syst. 151, 78–94 (2018)
- Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C., Yu, P.S.: A comprehensive survey on graph neural networks. CoRR, abs/1901.00596 (2019)
- Xu, M.: Understanding graph embedding methods and their applications, arXiv:2012.08019
- Zhang, D., Yin, J., Zhu, X., Zhang, C.: Network representation learning: a survey. CoRR, abs/1801.05852 (2018)

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## Network visualization

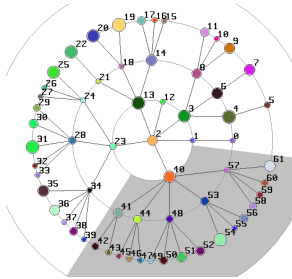
Different approaches:

- Fish eye
  - Centered on an actor
- Force driven visualization layouts
  - Kamada-Kawai (1989)
  - Fruchterman Reingold (1984)
    - Iterative algorithm



## Network visualization

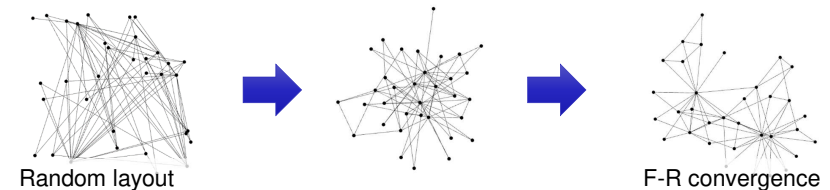
- Fish eye
  - Centered on an actor
  - With its neighbors
  - the neighbors of its neighbors..etc.



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## Network visualization

- Force driven visualization layout
  - Kamada-Kawai (1989)
  - Fruchterman Reingold (1984)
    - Iterative algorithm
    - Inspired by physics:
      - assigns forces among the set of edges and the set of nodes, based on their relative positions,
      - Use forces to attract pairs of endpoints of the graph's edges towards each other and repulsive forces like those of [electrically charged](#) particles to separate other pairs of nodes



## Network generation

- Introduction
- Networks representation, storage and visualization
- [Network analysis : measures and metrics](#)
- Models of network generation
- Node classification
- Community detection
- Influence/ link prediction
- Conclusion
- References
  
- Application with R

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## Social network analysis based on measures and metrics (SNA)

- Global indicators at network level by:
  - Number of nodes
  - Number of edges
  - Diameter
  - Density
  - Dyad and Triad census [Davis and Leinhardt, 1967]...
- Local indicators at node or edge level:
  - Number of neighbors → degree
  - Coefficient of clustering
  - Centrality measures (betweenness, closeness, eigenvector, [Freeman, 1979]
  - PageRank, Hits (authority, hub) [Brin et al, 1989, Kleinberg, 1999]
- Distance
  - Length of the shortest path

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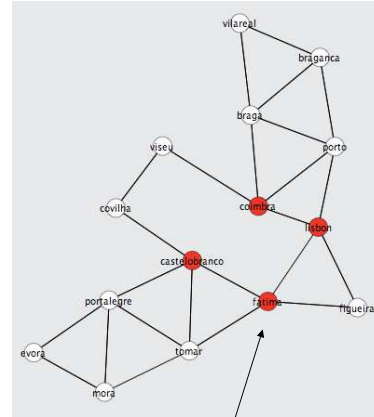
## Social network analysis (SNA)

Given the undirected graph  $G = (E, V)$  and its adjacency matrix  $A$

- Path from  $x$  to  $y$  : a sequence of vertices such that every consecutive pair of vertices in the sequence is connected by an edge in the network

$$(x, x_1), (x_1, x_2), \dots, (x_{k-1}, y)$$

- Length of a path: number of edges traversed along the path  
:  $k$



$K = 3$

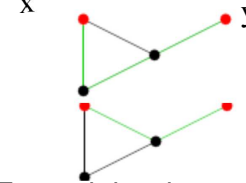
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## SNA based on measures and metrics (SNA)

- Number of paths of length  $r$  from  $i$  to  $j$  :  $A^r_{ij}$

$A^r_{ij}$  denotes the  $ij$  th element of the matrix  $A$  to the power  $r$

- Geodesic distance between two nodes  $x$  and  $y$ :  
length of a smallest path between  $x$  and  $y$  (non weighted network)



→ Length of the green path: 3

→ Geodesic distance between  $x$  and  $y$ : 2

For weighted networks: length of the path between  $x$  and  $y$  such that the sum of the weights of its constituent edges is minimized.

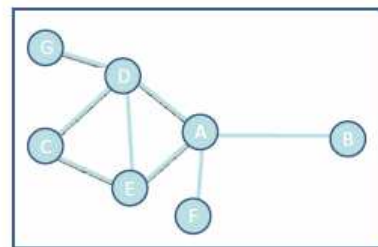
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## Social network analysis (SNA)

- Average path length (distance moyenne) : mean of the geodesic distances between the pairs of nodes

Weights of the edges

|   | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| A |   | 1 | 2 | 1 | 1 | 1 | 2 |
| B |   |   | 3 | 2 | 2 | 2 | 3 |
| C |   |   |   | 1 | 1 | 3 | 2 |
| D |   |   |   |   | 1 | 2 | 1 |
| E |   |   |   |   |   | 2 | 2 |
| F |   |   |   |   |   |   | 3 |
| G |   |   |   |   |   |   |   |



$$D(1)=8$$

$$D(2)=9$$

$$D(3)=4$$

$$L = (8 + 2 \times 9 + 3 \times 4) / (8 + 9 + 4) \quad L = 1.8$$

9

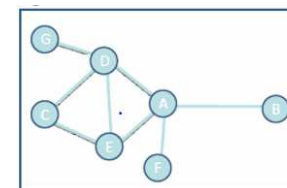
## Social network analysis (SNA)

- Diameter of  $G$ : maximum geodesic distance between the pairs of nodes



→

diameter = 2



→

diameter = ?

60

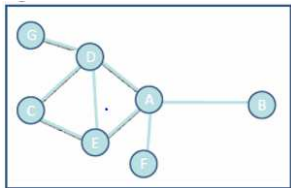
## Social network analysis (SNA)

- Diameter of G: maximum geodesic distance between the pairs of nodes



→

diameter = 2



→

diameter = 3

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## Social network analysis (SNA)

In practice in social networks:

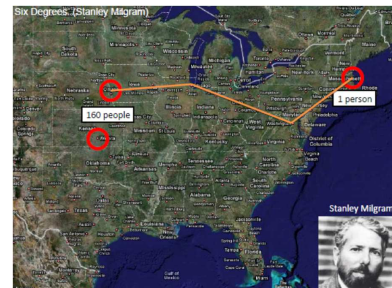
- Average distance and diameter of G are computed on the *biggest connected component*
- Short distances in real social networks:  
diameter  $\approx \ln(n)$   
(cf Wasserman98, Erdős-Rényi60, Milgram67 )

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## Social network analysis (SNA)

### Small word phenomena - Six degrees of separation Milgram Experiment (1967)

- Send 160 letters from Nebraska to Boston
- 44 letters arrived
- The average length of the chain between the sender and receiver was 5.5

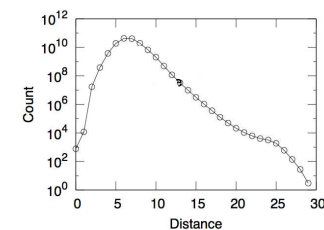
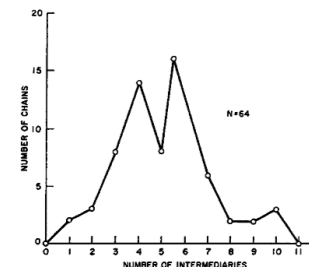


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## Social network analysis (SNA)

### • Small word phenomena

- Microsoft Internet messenger (IM) experiment (24000 active user accounts from 166 countries) by Jure Leskovec and Eric Horvitz (WWW 2008) :
  - avg. distance 6.6 & mean median 7
  - 4,74 on [Facebook](#)



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## Social network analysis (SNA)

- **Erdős number: Distance from him/her to Erdős in the coauthor graph**

- Paul Erdős  
a mathematician who published about 1500 papers

<http://www.oakland.edu/enp>

- 0 = Erdős
- 1 = Erdős collaborators
- 2 = Collaborators of Erdős collaborators
- 3 = Osmar Zaiane
- 4 = Christine Largeron



- Similarly, Kevin Bacon number  
<http://oracleofbacon.org/>

co-appearance in a movie

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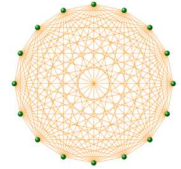
## Social network analysis (SNA)

- Density of  $G = (V, E)$  : probability of having a link in  $G$  (i.e. ratio of edges in  $G$  to the maximum possible number of edges)

If  $|V| = n$  and  $|E| = m$ ,  $\text{Density}(G) = \frac{2 \times m}{n \times (n-1)}$



→ density =  $(2 \times 4) / (4 \times 3)$



→ Low density in social network

- The maximum number of edges in an undirected graph is  $N(N-1)/2$ . It is called a complete graph.

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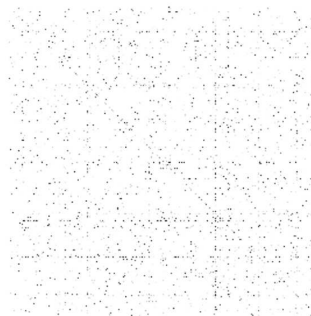
## Social network analysis (SNA)

- Low density in social network => sparse adjacency matrix filled with zeros!

Density of the matrix  $(E/N^2)$

WWW =  $1.51 \times 10^{-5}$

MSN IM =  $2.27 \times 10^{-8}$



Jure Leskovec, Stanford CS224W: Analysis of Networks

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## Social network analysis (SNA)

- Connexity:  $G$  is connected if there is a path between each pair of vertices in the network



- Connected component: set of vertices such that there is a path between each pair of vertices



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## Social network analysis (SNA)

### •Subgraph of G

A subgraph  $G_A = (A, E_A)$  of  $G=(V,E)$  induced by A is such that  $E_A$  is a subset of E corresponding to all the edges of E that connect pairs of vertices in A

$G$  has  $n(n-1)/2$  subgraphs of size 2 and  $n(n-1)(n-2)/6$  subgraphs of size 3

•**Dyad** : Each subgraph of size 2 comprises a pair of vertices, and the edge between them is either absent (D0) or present (D1)

**Dyad census** = (D0,D1) count of the no. of each type of dyad subgraph

$$D0 = n(n-1)/2 - m \text{ and } D1 = m$$

•**Triad census**: same analysis with subgraphs of size 3

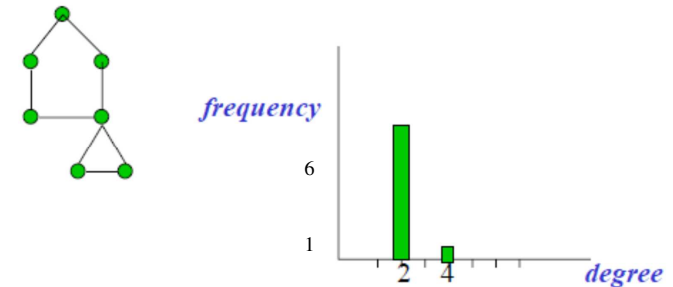
•Useful for community detection where dyad and triad are considered as the smallest community according to the size.

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## Social network analysis (SNA)

• Degree distribution of  $G : (N_0, N_1, \dots, N_{p-1})$  where

**Frequency** :  $N_k$  = the number of nodes with degree  $k$



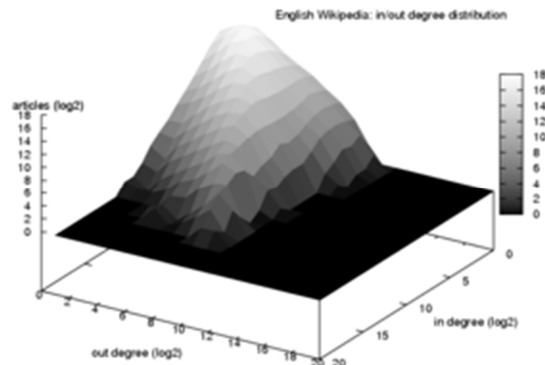
**Probability estimation** : fraction of vertex with degree  $k$

$$p_k = N_k/n$$

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## Social network analysis (SNA)

• Degree distribution in wikipedia



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## Social network analysis (SNA)

• Degree distribution models

– Bernoulli or random graph:

• each of  $n$  nodes is connected (or not) with independent probability  $p$  (or  $1 - p$ )

• Binomial distribution of degrees:

$$P(k) = Cn^k p^k (1-p)^{n-k-1}$$

• Binomial distribution can be approximate by Poisson in the limit of large  $n$

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## Social network analysis (SNA)

$P(k)$  : Probability that a node has  $k$  neighbors

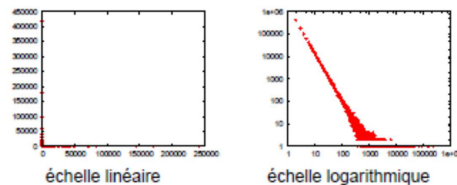
- Power law (or Pareto):

$$P(k) = k^{-\gamma}, \text{ where } \gamma \text{ is a constant}$$

a line in log log scale

- Degree distribution for scale free networks and www  
due to the rich get richer effect !

$$P(k) \sim k^{-\gamma}$$



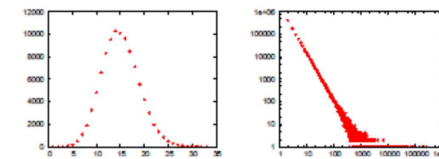
## Social network analysis (SNA)

- Homogeneous versus heterogeneous distribution

Homogeneous: normality (and exception)

Heterogeneous: no normality

in scale free network, www



Homogeneous

Heterogeneous (log-log scale)

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## Social network analysis (SNA)

- Local clustering coefficient of a node  $i$  with a neighbouring  $N_i$  of size  $k_i$  ( $>1$  – if  $k_i \leq 1$ ,  $C_i = 0$ )  
– for a directed graph:

$$C_i = \frac{|\{e_{jk} : v_j, v_k \in N_i, e_{jk} \in E\}|}{k_i(k_i - 1)}.$$

– for an undirected graph

$$C_i = \frac{2|\{e_{jk} : v_j, v_k \in N_i, e_{jk} \in E\}|}{k_i(k_i - 1)}.$$

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## Social network analysis (SNA)

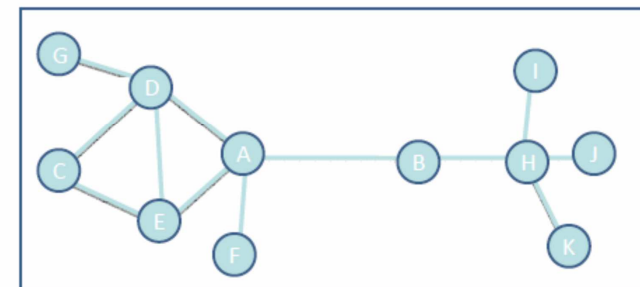
- Local clustering coefficient of a node  $i$  with a neighbouring  $N_i$  of size  $k_i$

$$C_i = \frac{2|\{e_{jk} : v_j, v_k \in N_i, e_{jk} \in E\}|}{k_i(k_i - 1)}.$$

$$C_A = 2/12 = 1/6$$

$$C_C = 2/2 = 1$$

$$C_E = 4/6 = 2/3$$

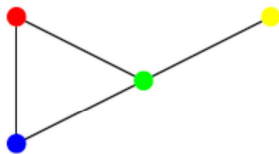


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## Social network analysis (SNA)

- Local clustering coefficient of a node  $i$  with a neighbouring  $N_i$  of size  $k_i$

$$C_i = \frac{2|\{e_{jk} : v_j, v_k \in N_i, e_{jk} \in E\}|}{k_i(k_i - 1)}.$$



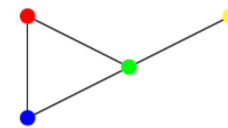
Clustering coefficient : 1, 1, 1/3, 0 : undefined

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## Social network analysis (SNA)

- Network average clustering coefficient (Watts and Strogatz):  
average of the local clustering coefficients of all the vertices having a degree  $\geq 2$

$$\bar{C} = \frac{1}{n} \sum_{i=1}^n C_i.$$



$$\bar{C} = (1+1+1/3)/3 = 0.77$$

$$\bar{C} = (1+1+1/3)/4 = 0.58$$

coefficients de clustering : 1, 1, 1/3, indéfini

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## Social network analysis (SNA)

- Network average clustering coefficient (Watts and Strogatz):

– Probability that the neighbours are connected between them → Community tendency

– In small-world network, the average clustering coefficient is significantly high

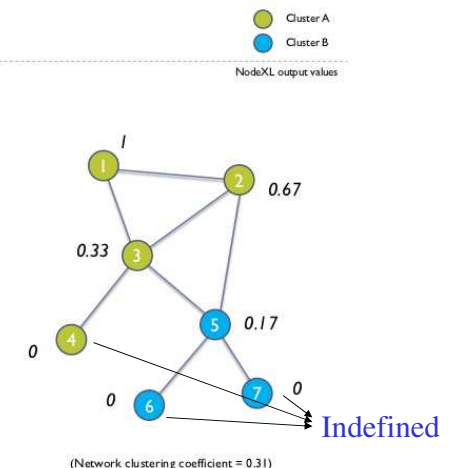
It can be compared to the expected clustering coefficient observed in a random graph having the same set of vertices.

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## Social network analysis (SNA)

### Clustering

- A node's *clustering coefficient* is the density of its *neighborhood* (i.e. the network consisting only of this node and all other nodes *directly* connected to it)
- E.g., node 1 to the right has a value of 1 because its neighbors are 2 and 3 and the neighborhood of nodes 1, 2 and 3 is perfectly connected (i.e. it is a 'clique')
- The clustering coefficient for an entire network is the average of all coefficients for its nodes
- Clustering algorithms** try to maximize the number of edges that fall within the same cluster (example shown to the right with two clusters identified)
- Clustering indicative of the presence of different (sub-)communities in a network



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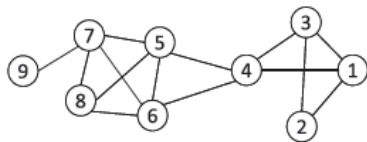
## Social network analysis (SNA) - centrality

**Degree Centrality** for a node x

$$C(x) = \text{degree}(x)$$

Normalized Degree Centrality for a node x

$$C(x) = \text{degree}(x) / (n-1) \text{ with } \text{Card}(V) = n$$



For node 1, degree centrality is 3;  
Normalized degree centrality is  
 $3/(9-1)=3/8$ .

4

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## Social network analysis (SNA) - centrality

Degree centrality for G:

$$C(G) = \sum_x \frac{(C_{\max} - C(x))}{(n-2)(n-1)} \quad (\text{Freeman, Wasserman})$$

where  $C_{\max}$  is the maximum value of the degree centrality observed on the nodes of G

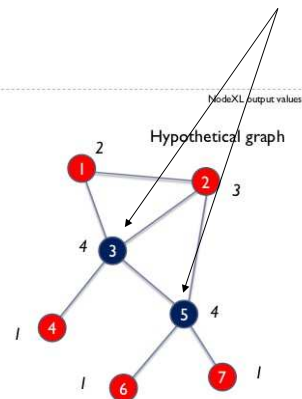
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## Social network analysis (SNA)

Degree Centrality for a node x : connectedness of x

### Degree centrality

- ▶ A node's (in-) or (out-)degree is the number of links that lead into or out of the node
- ▶ In an undirected graph they are of course identical
- ▶ Often used as measure of a node's degree of connectedness and hence also influence and/or popularity
- ▶ Useful in assessing which nodes are central with respect to spreading information and influencing others in their immediate 'neighborhood'



Nodes 3 and 5 have the highest degree (4).

▶ 18

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## Social network analysis (SNA)

Betweenness centrality for a vertex  $v_i$  (or an edge  $e$ )

Number of shortest paths from all vertices to all others that pass through that node (edge)

$$\text{Betweenness Centrality } C_B(v_i) = \sum_{v_s \neq v_i \neq v_t \in V, s < t} \frac{\sigma_{st}(v_i)}{\sigma_{st}}$$

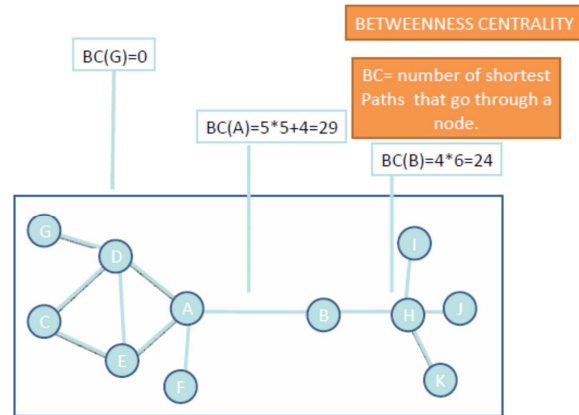
$\sigma_{st}$  : The number of shortest paths between s and t

$\sigma_{st}(v_i)$  : The number of shortest paths between s and t that pass  $v_i$

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## Social network analysis (SNA)

High betweenness centrality for a vertex (edge) means importance for the communication/diffusion



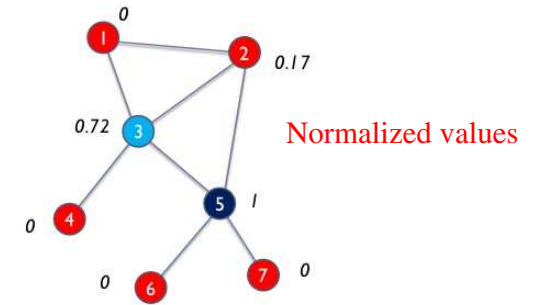
BC : values of the numerators only <> from  $C_B$  <sup>N=11</sup> <sub>85</sub>

## Social network analysis (SNA)

### Betweenness centrality

For a node or a link  
NodeXL output values

- ▶ The number of shortest paths that pass through a node divided by all shortest paths in the network
- ▶ Sometimes normalized such that the highest value is 1
- ▶ Shows which nodes are more likely to be in communication paths between other nodes
- ▶ Also useful in determining points where the network would break apart (think who would be cut off if nodes 3 or 5 would disappear)



Node 5 has higher betweenness centrality than 3

▶ 20

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## Social network analysis (SNA)

**Farness of a node:** sum of the lengths of the geodesic distance to the other nodes

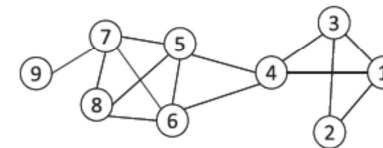
**Closeness centrality :** inverse of farness x (n-1)

$$C_C(v_i) = \left[ \frac{1}{n-1} \sum_{j \neq i} g(v_i, v_j) \right]^{-1} = \frac{n-1}{\sum_{j \neq i} g(v_i, v_j)}$$

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## Social network analysis (SNA)

### Closeness Centrality Example



| Node | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|------|---|---|---|---|---|---|---|---|---|
| 1    | 0 | 1 | 1 | 1 | 2 | 2 | 3 | 3 | 4 |
| 2    | 1 | 0 | 1 | 2 | 3 | 3 | 4 | 4 | 5 |
| 3    | 1 | 1 | 0 | 1 | 2 | 2 | 3 | 3 | 4 |
| 4    | 1 | 2 | 1 | 0 | 1 | 1 | 2 | 2 | 3 |
| 5    | 2 | 3 | 2 | 1 | 0 | 1 | 1 | 1 | 2 |
| 6    | 2 | 3 | 2 | 1 | 1 | 0 | 1 | 1 | 2 |
| 7    | 3 | 4 | 3 | 2 | 1 | 1 | 0 | 1 | 1 |
| 8    | 3 | 4 | 3 | 2 | 1 | 1 | 1 | 0 | 2 |
| 9    | 4 | 5 | 4 | 3 | 2 | 2 | 1 | 2 | 0 |

$$C_C(3) = \frac{9-1}{1+1+1+2+2+2+3+3+4} = 8/17 = 0.47,$$

$$C_C(4) = \frac{9-1}{1+2+1+1+1+1+2+2+3} = 8/13 = 0.62.$$

Node 4 is more central than node 3

6

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# Social network analysis (SNA)

## Farness of a node:

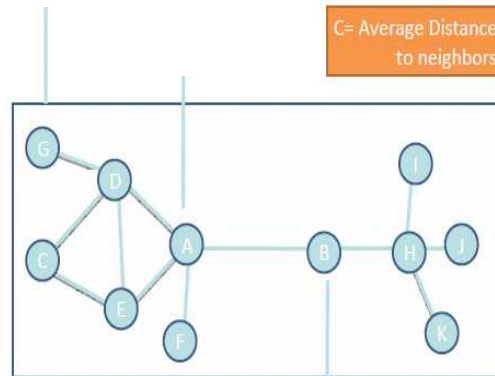
sum of the lengths of the geodesic distance to the other nodes

Closeness centrality of a node: inverse of farness x (n-1)

$$C(G) = (11-1) / (1*1+3*2+2*3+1*4+3*5) = 0.3125$$

CLOSENESS CENTRALITY

C = Average Distance to neighbors



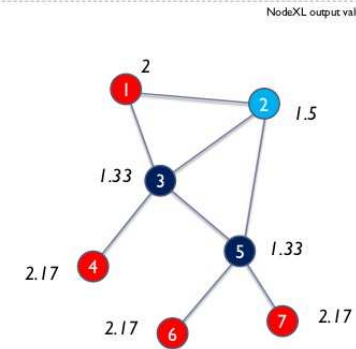
$$C(B) = (11-1) / (1*2+2*6+3*2) = 0.5$$

=11

# Social network analysis (SNA)

## Closeness centrality

- ▶ The mean length of all shortest paths from a node to all other nodes in the network (i.e. how many hops on average it takes to reach every other node)
- ▶ It is a measure of *reach*, i.e. how long it will take to reach other nodes from a given starting node
- ▶ Useful in cases where speed of information dissemination is main concern
- ▶ Lower values are better when higher speed is desirable



Fairness

Nodes 3 and 5 have the lowest (i.e. best) closeness, while node 2 fares almost as well

Note: Sometimes closeness is defined as the reciprocal of this value, i.e.  $1/x$ , such that higher values would indicate faster reach

▶ 21

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# Social network analysis (SNA)

## Eigenvector centrality

Consider the Adjacency Matrix  $A_{ij} = 1$  if node  $i$  is connected to node  $j$  and 0 otherwise.

Consider the eigenvalue problem:

$$Ax = \lambda x$$

Then the eigenvector centrality of a node is defined as:

$$x_i = \frac{1}{\lambda} \sum_{j \in V(i)} x_j = \frac{1}{\lambda} \sum_{j=1, n} A_{ij} x_j$$

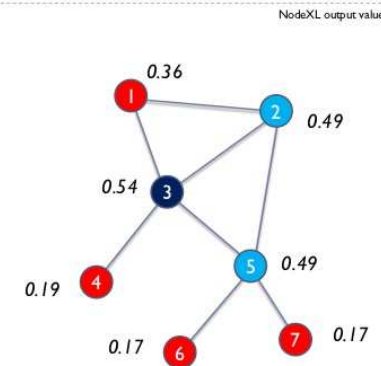
where  $\lambda$  is the largest eigenvalue associated with A.

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# Social network analysis (SNA)

## Eigenvector centrality

- ▶ A node's **eigenvector centrality** is proportional to the sum of the eigenvector centralities of all nodes directly connected to it
- ▶ In other words, a node with a high eigenvector centrality is connected to other nodes with high eigenvector centrality
- ▶ This is similar to how Google ranks web pages: links from highly linked-to pages count more
- ▶ Useful in determining who is connected to the most connected nodes



Node 3 has the highest eigenvector centrality, closely followed by 2 and 5

Note: The term 'eigenvector' comes from mathematics (matrix algebra), but it is not necessary for understanding how to interpret this measure

▶ 22

CNM Social Media Module – Giorgos Cheliotis (gcheliotis@nus.edu.sg)



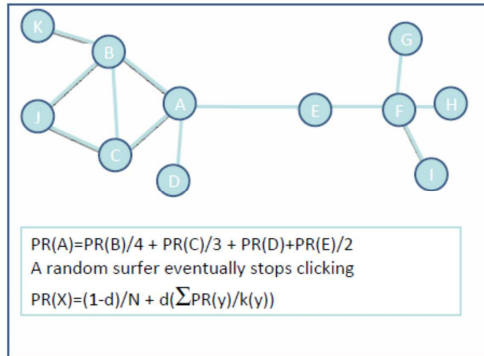
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# Social network analysis (SNA)

Page rank: particular case of eigenvector centrality  
Detailed in the sequel...

## PAGE RANK

PR=Probability that a random walker with interspersed jumps would visit that node.  
PR=Each page votes for its neighbors.



# Social network analysis (SNA)

## Interpretation of measures (1)

### Centrality measure

### Interpretation in social networks

#### ► Degree

How many people can this person reach directly?

#### ► Betweenness

How likely is this person to be the most direct route between two people in the network?

#### ► Closeness

How fast can this person reach everyone in the network?

#### ► Eigenvector

How well is this person connected to other well-connected people?

# Social network analysis (SNA)

## Interpretation of measures (2)

### Centrality measure

### Other possible interpretations...

#### ► Degree

In network of music collaborations: how many people has this person collaborated with?

#### ► Betweenness

In network of spies: who is the spy though whom most of the confidential information is likely to flow?

#### ► Closeness

In network of sexual relations: how fast will an STD spread from this person to the rest of the network?

#### ► Eigenvector

In network of paper citations: who is the author that is most cited by other well-cited authors?

# Outline

- Introduction
- Networks representation, storage and visualization
- Network analysis : measures and metrics
- **Models of network generation**
- Node classification
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- Influence/ link prediction
- Conclusion
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## Network generation

- Task:
  - Generation of networks
  - Comparison with random graphs
  - Analysis of the networks
  - Simulation/ Prediction
- Probabilistic models of networks in function of parameters:
  - Size of the network
  - Average degree of a vertex
  - etc

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## Network generation

Properties of canonical social network (www, etc)

- **Few** connected components:
  - often only 1 or a small number, indep. of network size
- **Small** diameter:
  - often a constant independent of network size (like 6)
  - or perhaps growing only logarithmically with network size or even shrink?
  - typically exclude infinite distances
- A **high** degree of clustering:
  - considerably more so than for a random network
  - in tension with small diameter
- A **heavy-tailed** degree distribution:
  - a small but reliable number of high-degree vertices
  - often of **power law** form

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## Network generation

### • Random network Models

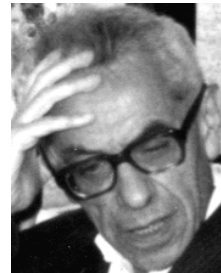
- Start with a set of  $n$  vertices
- Add  $m$  edges between them at random

### • Erdős-Rényi (ER) model (1959)

Alternative formulation

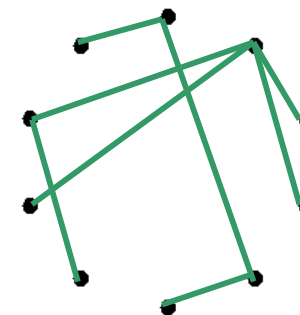
- $n$  nodes
- Select  $m$  links among  $(n) \cdot (n-1)/2$  with a probability  $p$
- Not scale free network, no high clustering structure

Pál Erdős  
(1913-1996)



Alfred Rényi

## Erdős-Rényi Model (1959)



Connect with  
probability  $p$

$$p=1/6$$

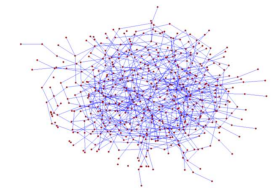
$$n=10$$

$$\langle k \rangle \sim 1.5$$



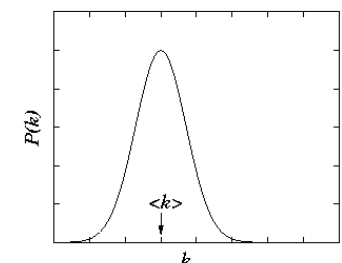
- Democratic
- Random

@ Han

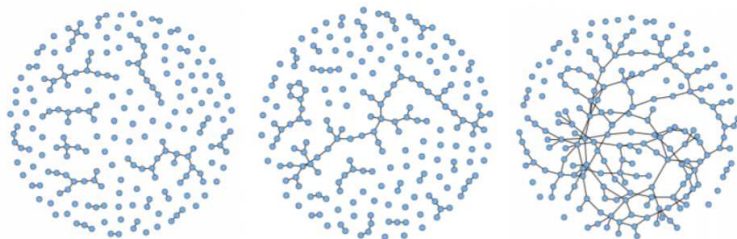


Pál Erdős  
(1913-1996)

Poisson distribution



## Erdős-Rényi Model (1959)



$p < p_c$

$p = p_c$

$p > p_c$

`igraph:erdos.renyi.game()`

Graph  $G(n, p)$ , for  $n \rightarrow \infty$ , critical value  $p_c = \frac{1}{n}$ .

- when  $p < p_c$  ( $k < 1$ ) there are no components with more than  $O(\ln(n))$  nodes, largest component is a tree
- when  $p = p_c$  ( $k = 1$ ) the largest component has  $O(n^{2/3})$  nodes
- when  $p > p_c$  ( $k > 1$ ) gigantic component has all  $O(n)$  nodes

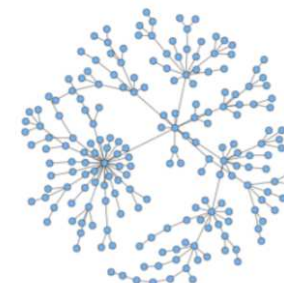
@[https://cs.hse.ru/data/2015/04/30/1098190683/3.\\_Graph\\_Models.pdf](https://cs.hse.ru/data/2015/04/30/1098190683/3._Graph_Models.pdf) 101

## Network generation

- **Preferential attachment model** (Barabasi, 1999): rich get richer
  - Add the node one by one
  - Add the links to the nodes already connected according to their degree (ie with a preference for the nodes which have already a high degree)



Rich get richer !



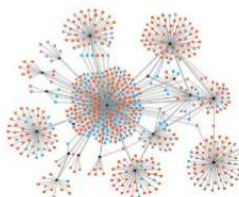
## Network generation

### Preferential Attachment

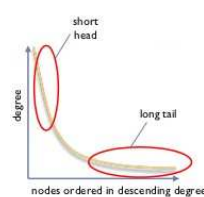
A property of some networks, where, during their evolution and growth in time, a the great majority of new edges are to nodes with an already high degree; the degree of these nodes thus increases disproportionately, compared to most other nodes in the network

- ▶ The result is a network with few very highly connected nodes and many nodes with a low degree
- ▶ Such networks are said to exhibit a *long-tailed* degree distribution
- ▶ And they tend to have a small-world structure!

(so, as it turns out, transitivity and strong/weak tie characteristics are not necessary to explain small world structures, but they are common and can also lead to such structures)



Example of network with preferential attachment



Sketch of long-tailed degree distribution

## Network generation

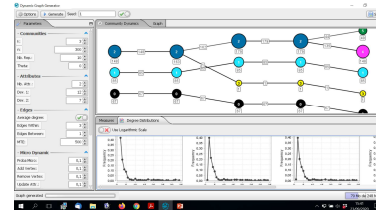
### Models of network generation

- **Random graph (Erdős-Rényi model):**
  - Gives few components and small diameter
  - does not give high clustering and heavy-tailed degree distributions
  - is the mathematically most well-studied and understood model
- **Watts-Strogatz model:**
  - gives few components, small diameter and high clustering
  - does not give heavy-tailed degree distributions
- **Scale-free Network:**
  - gives few components, small diameter and heavy-tailed distribution
  - does not give high clustering
- **Hierarchical network** (part of the scale-free model):
  - few components, small diameter, heavy-tailed, **high clustering**

# Network generation

## Models of network generation

- **Affiliation network:**
  - models group-actor formation i.e. bipartite graph (two modes)
- **Attributed graph (information network) with community structure: graph in which the nodes are described by attributes and organized in communities**
  - Dang et al, 2012
  - Largeron et al. PlosOne2015, Kais2017



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# Outline

- Introduction
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  - **Node Classification**
  - Community detection
  - Influence/ link prediction
  - Conclusion
  - References
- 
- Application with R

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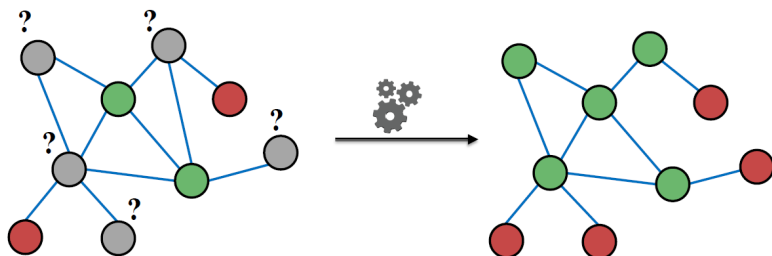
# Node classification

**Aim :** Given a graph with labels on some nodes, predict the label of unlabeled nodes.

**Input:** A graph  $G=(V,E)$  with labels (class membership) on some nodes

**Output:** The labels of unlabeled nodes

**Semi-supervised task**

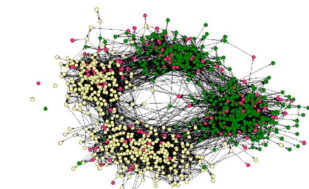
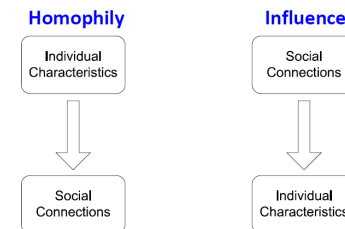


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# Node classification

**Hypothesis: similar nodes are connected**

- **Homophily effect:** Tendency of individuals to associate and bond with similar others
- **Social influence:** Social connections influence the individual characteristics



Node color = interests (sports, arts, etc.)  
(Easley and Kleinberg, 2010)

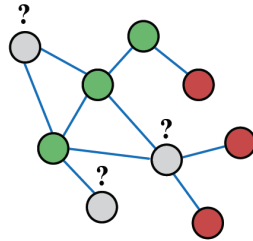
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# Node classification

## Methods:

With homophily assumption

- K nearest neighbors
- Iterative classification
- Random walk
- Label propagation
- Node embeddings -> SVM,..
- Graph neural networks
  - Graph attention network (GAT) [Velickovic et al., 2018]
  - Graph Convolution Networks [Kipf and Welling, 2017]
  - GraphSage [Hamilton et al., 2017]



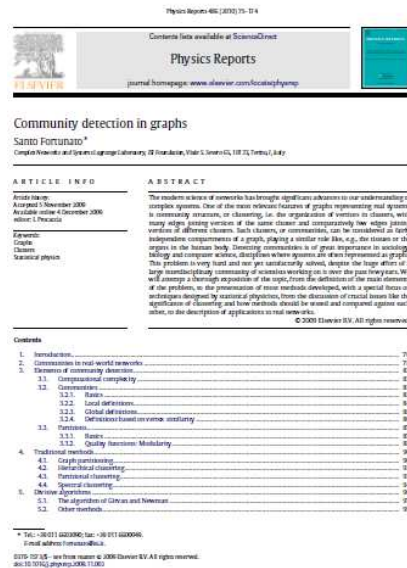
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# Outline

- Introduction
  - Networks representation, storage and visualization
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  - Models of network generation
  - Node Classification
  - **Community detection**
  - Influence/ link prediction
  - Conclusion
  - References
- 
- Application with R

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# Community detection



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# Community detection

- Santo Fortunato,  
**Darko Hric**:  
Community detection  
in networks: A user  
guide. **CoRR**  
[abs/1608.00163](https://arxiv.org/abs/1608.00163)  
(2016)

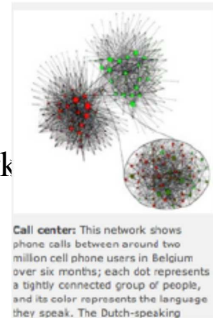
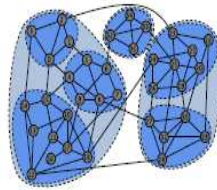


## Community detection

- Aim
  - Automatic detection of groups of nodes
    - Input: social network i.e.  $G = (V, E)$
    - Output: community membership of (some) actors

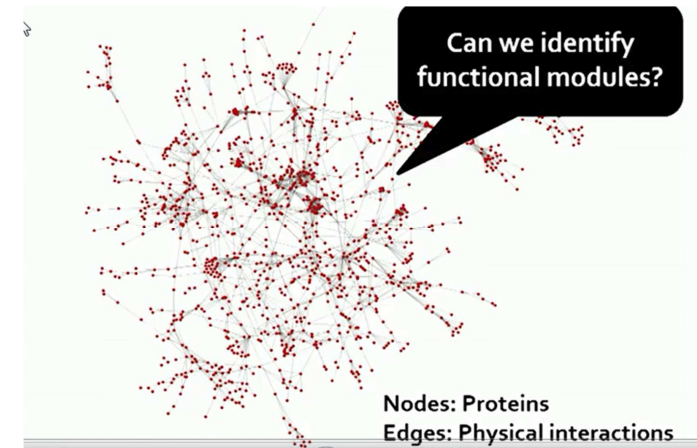
### Unsupervised task

- Applications
  - Understanding the interactions between people
  - Visualizing and navigating in huge network
  - Forming the basis for other tasks such as data mining, improving of systems (retrieval, recommendation, etc)
  - Detection of specific communities



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## Community detection



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## Community detection

- What is a community?
  - No formal definition
    - A clique: subgraph completely connected ?
    - A subgraph highly connected ?
    - A subset of actors among whom there are relatively strong, direct, intense, frequent or positive ties. (Wasserman1994)
- Community structure of small words networks
  - Low global density but parts of the graph with high density (the communities)
  - Short length of the paths (small world)

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## Community detection

- Classical community detection :
  - Partitioning the set  $V$  of vertices into  $r$  communities  $P = \{C_1, \dots, C_r\}$

$$\begin{aligned}
 & \bullet \bigcup_{k \in \{1, \dots, r\}} C_k = V \\
 & \bullet C_k \cap C_l = \emptyset, \forall 1 \leq k < l \leq r \\
 & \bullet C_k \neq \emptyset, \forall k \in \{1, \dots, r\}
 \end{aligned}$$

such that the vertices in the same community **have strong connexions between** them and low connexions with the others

- Usual definition based on the strenght of the links but
  - Ambiguous: what means strongly connected?
  - Not universal

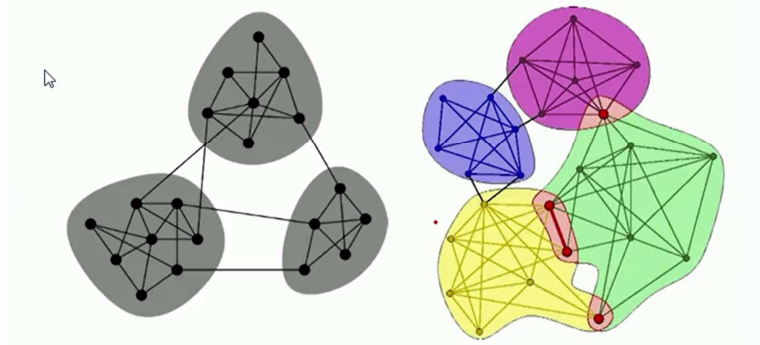
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## Community detection

- Overlapping communities (Cfinder, Clique Percolation Method, Palla2005, Xie2013)

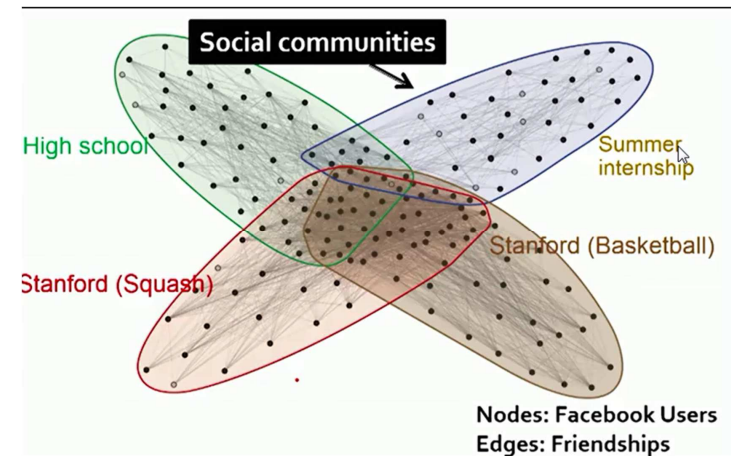
A vertex can belong to several communities

### ■ Non-overlapping vs. overlapping communities



## Community detection

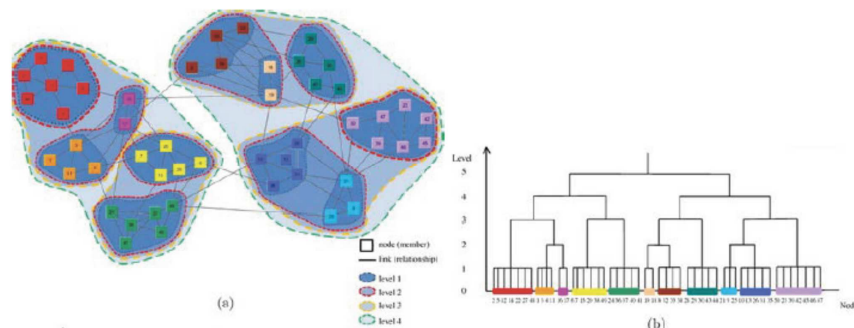
- Overlapping communities (Cfinder, Clique Percolation Method, Palla et al.)



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## Community detection

- Hierarchical communities: sequence of partitions [Girwan & Newman2002, Yang2013]

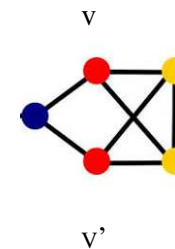


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## Community detection

- Other definitions
  - Based on structural equivalence [Wasserman1994]

Two nodes belong to the same community if they have the same neighbours.

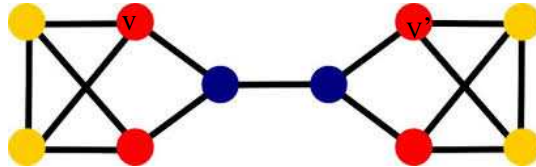


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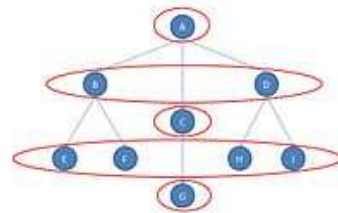
## Community detection

- Other definitions
  - Based on automorphic equivalence [Borgatti et al. 92] (isomorphic)
 

Two nodes share the same set of relationships to subsets of nodes that are isomorphic, but not necessarily the same set of neighbors.



five automorphic equivalence classes: {A}, {B, D}, {C}, {E, F, H, I}, and {G}

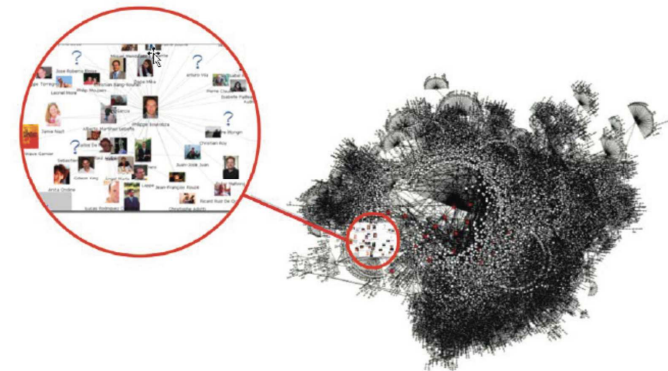


## Community detection

- Other definitions
  - Based on the leadership [Shah2010, Rabbaghy2010, Kanawati2014]
 

Two nodes belong to the same community if they follow the same leader
  - See also community discovery /search or local community detection:
 

Identify the community of a given node without seeing the whole network ..



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## Community detection

Community detection versus clustering  
Same problem but based on structure i.e. the links between the vertices than on the attributes describing the elements

Community detection and clustering share common goal of grouping objects, but differ in fundamental ways:

| Clustering       | Community Detection  |
|------------------|----------------------|
| Vectors          | Vertices             |
| Similarity       | Connectivity         |
| Metric structure | Relational structure |

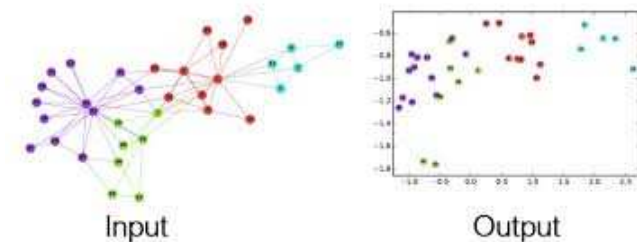
→ Graph embedding (Cai2019, Hamilton 2019)

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## Community detection

Graph embedding (Cai2019, Hamilton 2019)

- project nodes, edges, or parts of a graph into a vector space (of lower dimension) whilst maximally preserving properties like graph structure



- Node2vec, LLE, Deepwalk, HOPE, SDNE etc  
<https://github.com/palash1992/GEM>

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## Community detection

- Community validation: difficult like for clustering since definition of a community is subjective
  - Quality functions also used to build the partition
    - Normalized cut (Shi,2000)
    - Modularity (Newman and Girvan, 2004)
  - Ground truth is not always available: non supervised task

Reihaneh Rabbany, Mansoureh Takaffoli, Justin Fagnan, Osmar R. Zaiane, Ricardo J. G. B. Campello: Communities validity: methodical evaluation of community mining algorithms. Soc. Netw. Anal. Min. 3(4): 1039-1062 (2013)

Reihaneh Rabbany, Osmar R. Zaiane: Evaluation of Community Mining Algorithms in the Presence of Attributes. PAKDD Workshops 2015: 152-163

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## Community detection algorithms

- Min-cut [Goldberg and Tarjan1988] Ratio et normalized cuts [Wei et Chen 1989, Shi et Malik, 2000]
- Edge betweenness [Girvan et Newman,2001]
- Spectral clustering [Ng et al.,2002]
- Modularité [Newman et Girvan2004, Blondel et al. 2008] Walktrap [Pons et Latapy,2005]
- Conductance [Leskovec2009] [Rosvall et al. 2009]
- Stochastic Block model [Holland et al. 1983, Snijders et Nowicki,1997]
- Mixed Membership block model [Airoldi et al. 2008]
- Density based extraction [Zhao , Levina Zhou, 2012]
- etc.

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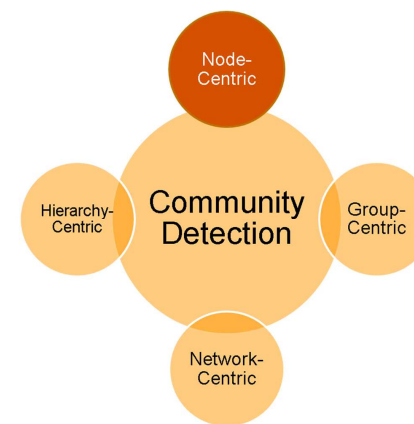
## Community detection algorithms

4 categories (not exclusive):

- **Node-Centric** Community
  - Each node in a group satisfies certain properties
- **Group-Centric** Community
  - Consider the connections within a group as a whole. The group has to satisfy certain properties without zooming into node-level
- **Network-Centric** Community
  - Partition the whole network into several disjoint sets
- **Hierarchy-Centric** Community
  - Construct a hierarchical structure of communities

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## Community detection algorithms



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## Community detection algorithms

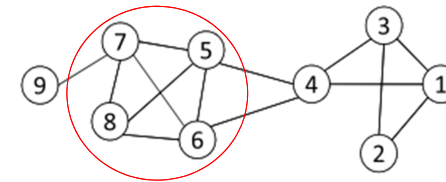
### Node centric approaches

- used in traditional social network analysis
- Nodes satisfy different properties
  - Complete Mutuality
    - cliques
  - Reachability of members
    - k-clique, k-clan, k-club
  - Nodal degrees
    - k-plex, k-core
  - Relative frequency of Within-Outside Ties
    - LS sets, Lambda sets

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## Community detection

- **A clique:** i.e. a maximum complete subgraph in which all nodes are adjacent to each other



- NP-hard to find the maximum clique in a network (Bomze 1999)
- Straightforward implementation to find cliques is very expensive in time complexity (Bron 1973)

## Community detection

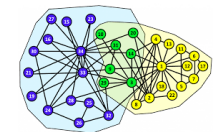
- **Clique:** definition of a community too strict
  - Reachability of members: k-clique [Alba, 1973], k-clan, k-club [Mokken, 1979]
  - Vertex degree: k-plex, k-core [Seidman et al. 1978, 1983]
  - Internal versus external cohesion: LS-sets [Luccio and Sami 1969] strong community [Radicchi et al. 2004], Lambda sets [Borgatti et al. 1990]

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## Community detection

### Clique Percolation Method (CPM) [Palla et al., 2005]

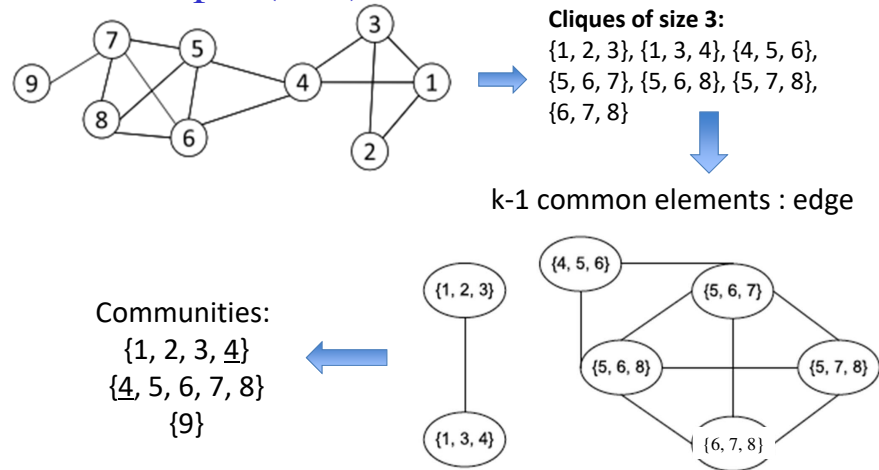
- Clique is a very strict definition, unstable
- Normally use cliques as a **core** or a **seed** to find larger communities
- CPM is a method to find **overlapping** communities
  - **Input**
    - A parameter  $k$ , and a network
  - **Procedure**
    - Find out all cliques of size  $k$  in a given network
    - Construct a **clique graph**. Two cliques are adjacent if they share  $k-1$  nodes
    - Each **connected** component in the clique graph forms a community
- Extension of CPM: directed graphs - CPMd, Weighted graphs - CPMw



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## Community detection

### CPM Example (k=3)



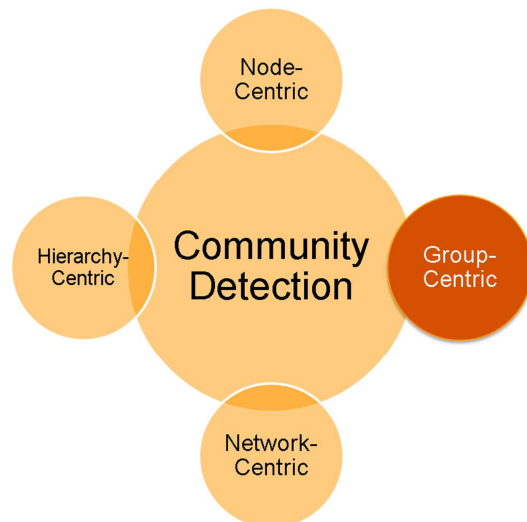
## Community detection algorithms

### Node centric approaches

- **Limitations:**
  - Too strict, but can be used as the core of a community
  - Not scalable, commonly used in network analysis with small-size network

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## Community detection algorithms



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## Community detection algorithms

### Group centric approaches

- The group-centric criterion requires the whole group to satisfy a certain condition
  - E.g., the group density  $\geq$  a given threshold
- A subgraph  $G_s(V_s, E_s)$  is a  $\gamma$ -dense quasi-clique if

$$\frac{2|E_s|}{|V_s|(|V_s| - 1)} \geq \gamma$$

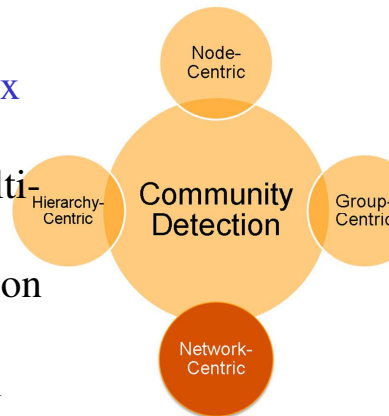
where the denominator is the maximum number of degrees.

- A similar strategy to that of cliques can be used
  - Sample a subgraph, and find a maximal  $\gamma$ -dense quasi-clique (say, of size  $|V_s|$ )
  - Remove nodes with degree less than the average degree  $|V_s|\gamma \leq \frac{2|E_s|}{|V_s|-1}$

# Community detection algorithms

Network centrix approaches:

- (1) Clustering based on vertex similarity
- (2) Latent space models (multi-dimensional scaling )
- (3) Block model approximation
- (4) Spectral clustering
- (5) Modularity maximization

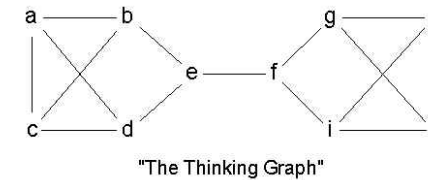


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# Community detection algorithms

Clustering based on vertex similarity

- Vertex similarity is defined in terms of the similarity of their neighborhood
- Structural equivalence:** two nodes are structurally equivalent iff they are connecting to the same set of actors



- Structural equivalence is too strict for practical use.

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# Community detection algorithms

Clustering based on vertex similarity

- Alternative strategy:
  - Define the similarity measure between vertex (Jaccard, Hamming, Cosinus,..)
  - Apply a clustering algorithm ( k-means, hierarchical algorithms, etc.)

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# Community detection algorithms

Vertex similarity

- Jaccard Similarity  $Jaccard(v_i, v_j) = \frac{|N_i \cap N_j|}{|N_i \cup N_j|}$
- Cosine similarity  $Cosine(v_i, v_j) = \frac{|N_i \cap N_j|}{\sqrt{|N_i| \cdot |N_j|}}$

- Hamming distance between rows of the adjacency matrix

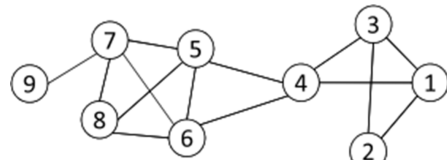
$$d(v_i, v_j) = \sum_l A_{il} \oplus A_{jl} \quad \oplus = \text{XOR}$$

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# Community detection algorithms

## Vertex similarity

- Jaccard Similarity  $Jaccard(v_i, v_j) = \frac{|N_i \cap N_j|}{|N_i \cup N_j|}$
- Cosine similarity  $Cosine(v_i, v_j) = \frac{|N_i \cap N_j|}{\sqrt{|N_i| \cdot |N_j|}}$



$$Jaccard(4, 6) = \frac{|\{5\}|}{|\{1, 3, 4, 5, 6, 7, 8\}|} = \frac{1}{7}$$

$$cosine(4, 6) = \frac{1}{\sqrt{4 \cdot 4}} = \frac{1}{4}$$

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# Community detection algorithms

## Clustering based on Latent Space Models (cf PCA - M1 W1)

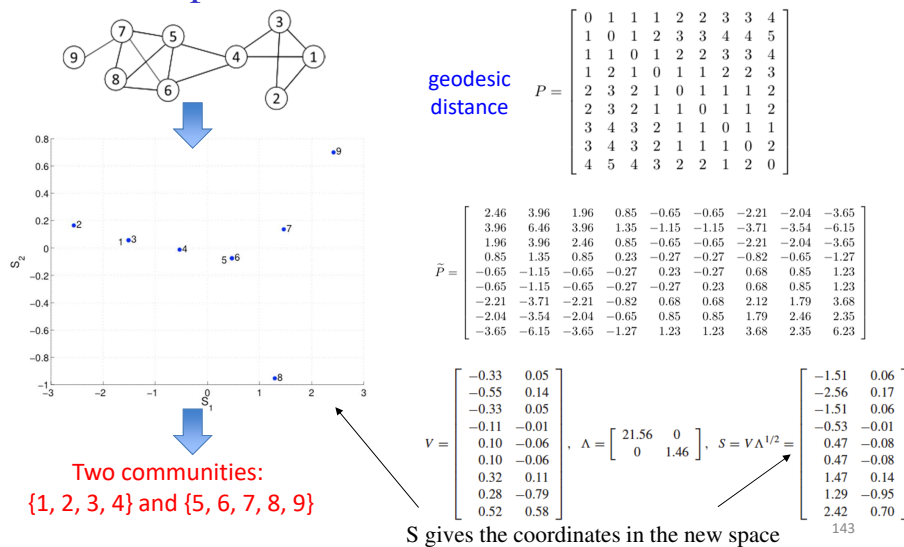
Map nodes into a low-dimensional space such that the proximity between nodes based on network connectivity is preserved in the new space, then apply k-means clustering

- Multi-dimensional scaling (MDS)
    - Given a network, construct a proximity matrix P representing the pairwise distance between nodes (e.g., geodesic distance)
    - Let  $S \in R^{n \times d}$  denote the coordinates of nodes in the low-dimensional space
- Centered matrix**  $SS^T \approx -\frac{1}{2}(I - \frac{1}{n}\mathbf{1}\mathbf{1}^T)(P \circ P)(I - \frac{1}{n}\mathbf{1}\mathbf{1}^T) = \tilde{P}$
- Objective function:**  $\min \|SS^T - \tilde{P}\|_F^2$
  - Solution:**  $S = V \cdot \sqrt{\Lambda}$
  - V is the top  $\ell$  eigenvectors of  $\tilde{P}$ , and  $\Lambda$  is a diagonal matrix of top eigenvalues  $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_\ell)$

- Related to embedding based methods : Node2vec or Deepwalk etc

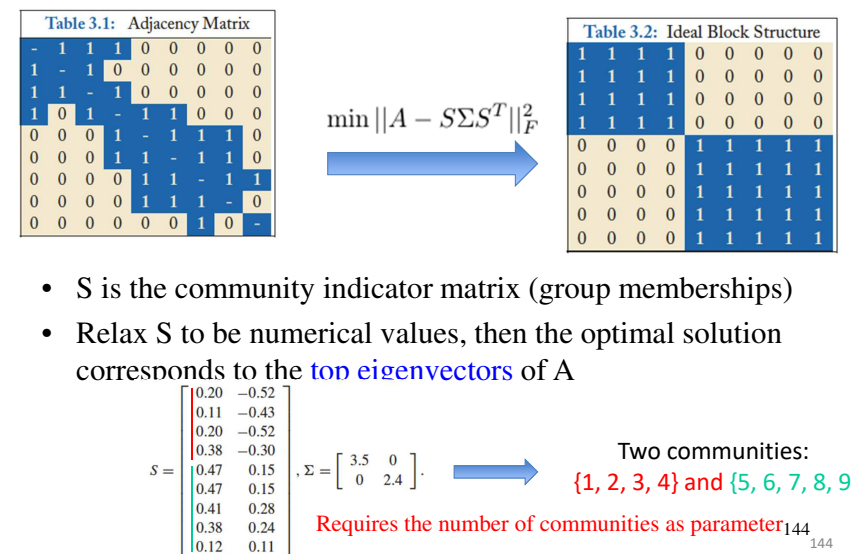
# Community detection algorithms

## MDS Example



# Community detection algorithms

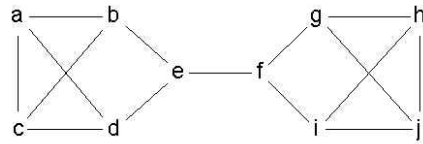
## Block Models approaches



# Community detection algorithms

## Spectral clustering: Min Cut, ratio, normalized cut

- Hypothesis: Most interactions are within group whereas interactions between groups are few
- community detection → **minimum cut problem**
- Cut**: A partition of vertices of a graph into two disjoint sets
- Minimum cut problem**: find a graph partition such that the number of edges between the two sets is minimized



"The Thinking Graph"

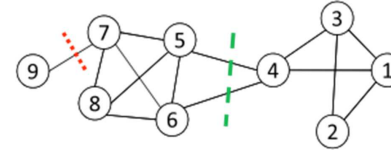
1 cut : Partition  
**C1** : a,b,c,d,e  
**C2** : f, g, h, i, j

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# Community detection algorithms

## Spectral clustering: Min Cut, ratio, normalized cut

- Cut**: A partition of vertices of a graph into two disjoint sets
- Minimum cut problem**: find a graph partition such that the number of edges between the two sets is minimized
- often** returns an **imbalanced** partition, with one set being a singleton, e.g. node 9



2 Partitions

**1 cut** :  $\Pi_1$  {9}, {1,2,3,4,5,6,7,8}  
**Imbalanced**

**2 cut** :  $\Pi_2$  {5, 6, 7, 8, 9}, {1,2,3,4}

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# Community detection algorithms

## Ratio Cut & Normalized Cut

- Change the objective function to consider community size

$$\text{Ratio Cut}(\pi) = \frac{1}{k} \sum_{i=1}^k \frac{\text{cut}(C_i, \bar{C}_i)}{|C_i|},$$

$$\text{Normalized Cut}(\pi) = \frac{1}{k} \sum_{i=1}^k \frac{\text{cut}(C_i, \bar{C}_i)}{\text{vol}(C_i)}$$

$C_i$ : a community  
 $|C_i|$ : number of nodes in  $C_i$   
 $\text{vol}(C_i)$ : sum of degrees of nodes in  $C_i$

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# Community detection algorithms

## Ratio Cut & Normalized Cut Example

**For partition in red:  $\pi_1$**

$$\text{Ratio Cut}(\pi_1) = \frac{1}{2} \left( \frac{1}{1} + \frac{1}{8} \right) = 9/16 = 0.56$$

$$\text{Normalized Cut}(\pi_1) = \frac{1}{2} \left( \frac{1}{1} + \frac{1}{27} \right) = 14/27 = 0.52$$

**For partition in green:  $\pi_2$**

$$\text{Ratio Cut}(\pi_2) = \frac{1}{2} \left( \frac{2}{4} + \frac{2}{5} \right) = 9/20 = 0.45 < \text{Ratio Cut}(\pi_1)$$

$$\text{Normalized Cut}(\pi_2) = \frac{1}{2} \left( \frac{2}{12} + \frac{2}{16} \right) = 7/48 = 0.15 < \text{Normalized Cut}(\pi_1)$$

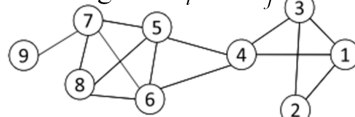
Both ratio cut and normalized cut prefer a **balanced partition**

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# Community detection algorithms

## Modularity maximization (Newman,2004)

- Modularity measures the strength of a community partition by taking into account the degree distribution
- H0:** Given a network with  $m$  edges,  $n$  nodes and same degree distribution as  $G$ , the expected number of edges between two nodes with degrees  $d_i$  and  $d_j$  is  $d_i d_j / 2m$



The expected number of edges between nodes 1 and 2 is  
 $3 \cdot 2 / (2 \cdot 14) = 3/14$

Modularity: 
$$Q = \frac{1}{2m} \sum_{\ell=1}^k \sum_{i \in C_\ell, j \in C_\ell} (A_{ij} - d_i d_j / 2m)$$

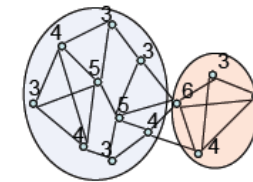
- Varies between -1 and 1: a larger value indicates a good community structure on the network

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# Community detection algorithms

## Modularity

### Quality of a partition of a network in communities



$$\frac{\text{\# edges in communities} - \text{expected \# edges in communities}}{\text{total number of edges}} = \frac{(15 + 6) - (11.56 + 2.56)}{25} = 0.275$$

Expected 11.56 2.56  
 Observed 15 6

$m$  nodes,  $d_i$  degree of node  $i$

expected # edges between node  $i$  and node  $j = (d_i d_j) / (2m)$

expected # edges in community  $\sum_{i,j \in c} (d_i d_j) / (2m)$

[Newman, 2004]

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# Community detection algorithms

## Properties of modularity:

- Between [-1, 1]
- Modularity = 0 if all nodes are clustered into one group
- Can automatically determine optimal number of groups

### Limits

- Optimizing modularity is NP hard( Arenas2008)
- Field of view limit and resolution limit of modularity (Fortunato2007, Marcotorchino2014)
  - Modularity maximization might return a community consisting multiple small modules or fails to detect well defined small communities

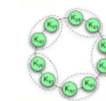
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# Community detection algorithms

## Resolution limit of modularity (Fortunato2007, Marcotorchino2014)

Modularity maximization fails to detect small well defined communities

Example: Ring of  $\alpha = 30$  cliques of size  $\beta = 5$  with  $Q_{single} = 0.876$ ,  $Q_{pairs} = 0.888$  (@Tarissan et al.)

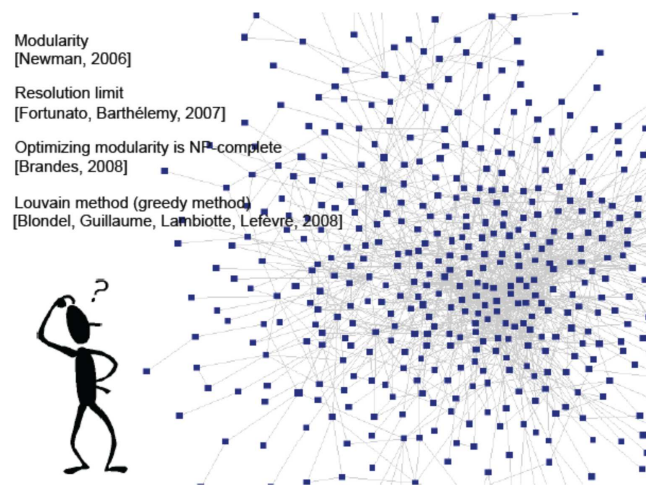


Tendency to favor large communities

Solutions: [Arenas et al. 2008], [Reichardt et Bornholdt, 2006]

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## Community detection algorithms



@Blondel 153

## Louvain algorithm

Initially **every node** forms a community

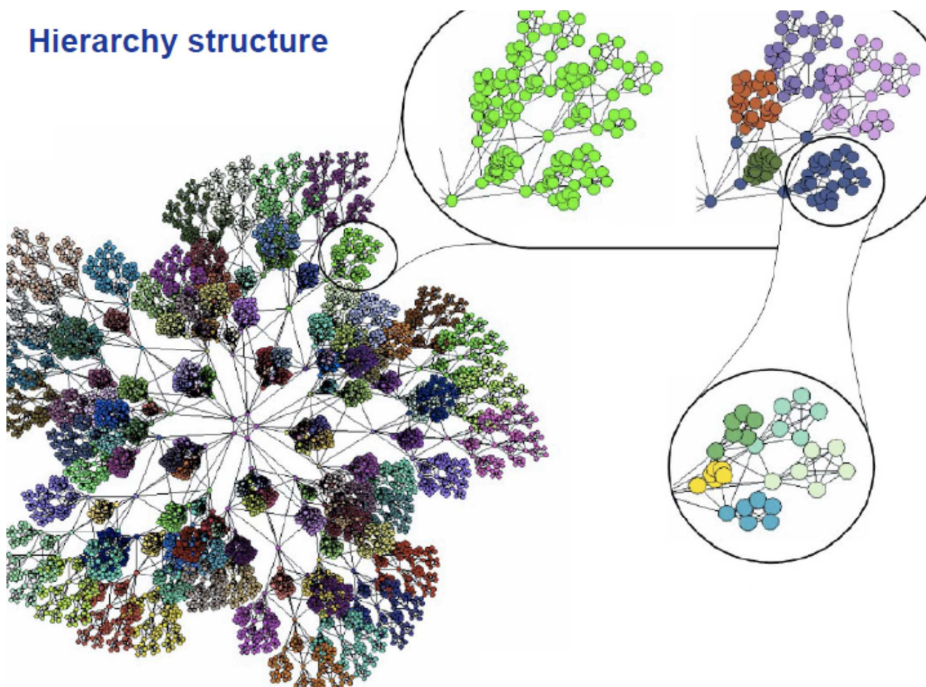
- repeat iteratively for all nodes  $i$ :
  - remove  $i$  from its community;
  - insert  $i$  in a neighboring community of  $i$  so as to maximize modularity (local greedy approach);
- stop when a local maximum is attained.

Construct the graph of communities and **repeat**

This way we compute **communities of communities**, etc

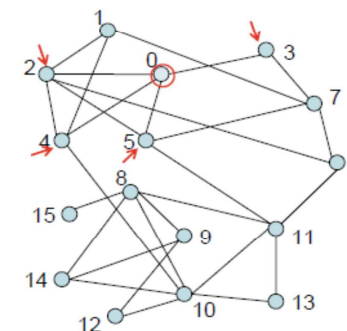
@Blondel 154

## Hierarchy structure



## Louvain algorithm

### An example

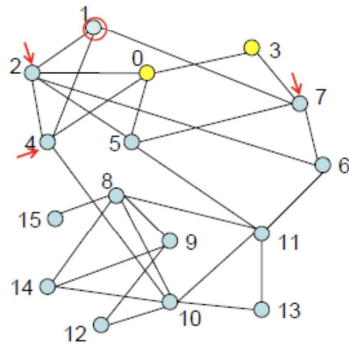


Pass 1 – Iteration 1  
Each node belongs to an atomic community

Ex: J-L Guillaume

## Louvain algorithm

### An example

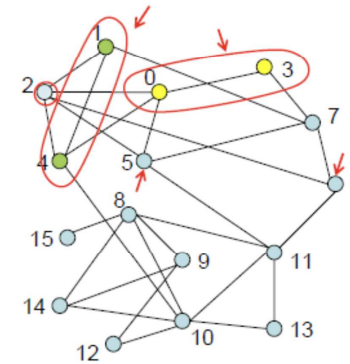


Pass 1 – Iteration 1  
insert 0 in  $c[3]$

@Guillaume<sup>157</sup>

## Louvain algorithm

### An example

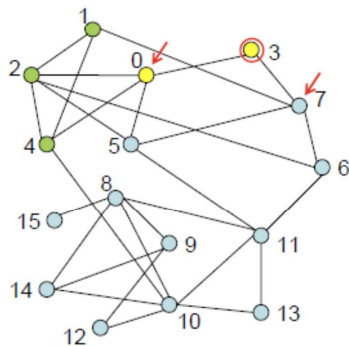


Pass 1 – Iteration 1  
insert 0 in  $c[3]$   
insert 1 in  $c[4]$

@Guillaume<sup>158</sup>

## Louvain algorithm

### An example

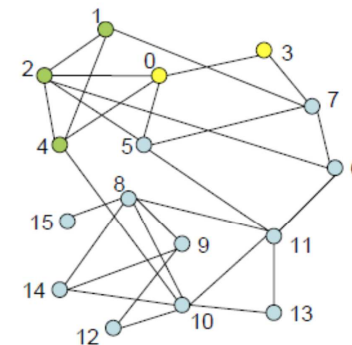


Pass 1 – Iteration 1  
insert 0 in  $c[3]$   
insert 1 in  $c[4]$   
insert 2 in  $c[1,4]$

@Guillaume<sup>159</sup>

## Louvain algorithm

### An example

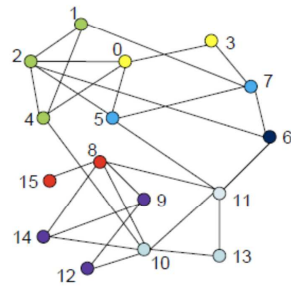


Pass 1 – Iteration 1  
insert 0 in  $c[3]$   
insert 1 in  $c[4]$   
insert 2 in  $c[1,4]$   
insert 3 in  $c[0]$

@Guillaume<sup>160</sup>

## Louvain algorithm

### An example

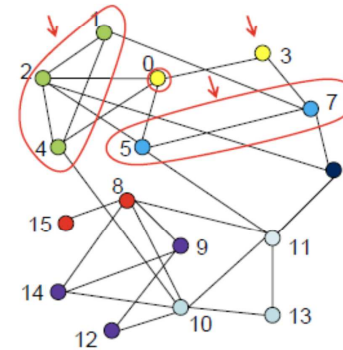


Pass 1 – Iteration 1  
 insert 0 in c[3]  
 insert 1 in c[4]  
 insert 2 in c[1,4]  
 insert 3 in c[0]  
 insert 4 in c[1]  
 insert 5 in c[7]  
 insert 6 in c[11]  
 insert 7 in c[5]  
 insert 8 in c[15]  
 insert 9 in c[12]  
 insert 10 in c[13]  
 insert 11 in c[10,13]  
 insert 12 in c[9]  
 insert 13 in c[10,11]  
 insert 14 in c[9,12]  
 insert 15 in c[8]

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 @Guillaume

## Louvain algorithm

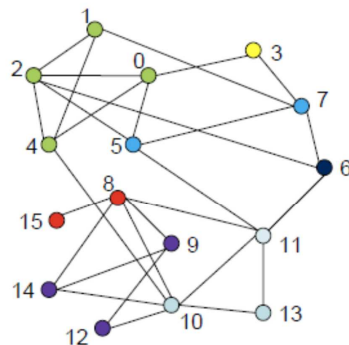
### An example



Pass 1 – Iteration 2

## Louvain algorithm

### An example

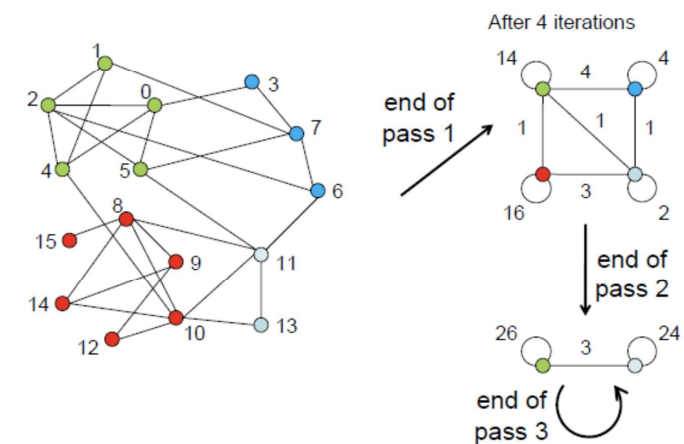


Pass 1 – Iteration 2  
 insert 0 in c[4]  
 ...

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 @Guillaume

## Louvain algorithm

### An example



## Louvain algorithm

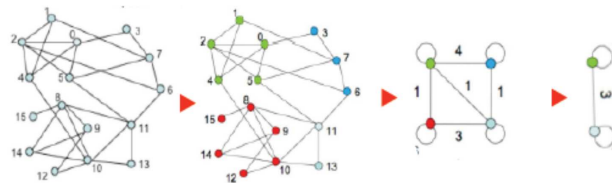
- 1-Initialization
- 2- Repeat alternatively step A and B
  - A- Reaffectation
  - B- New graph construction

Initially **every node** forms a community.

For every node  $i$ , insert node  $i$  in a neighboring community that maximizes the resulting modularity gain.

Repeat until a local maximum is attained.

Construct the resulting **network of communities** and **repeat** the construction on the network of communities.



## Louvain algorithm

**Hierarchy of communities**

**Simple** to implement

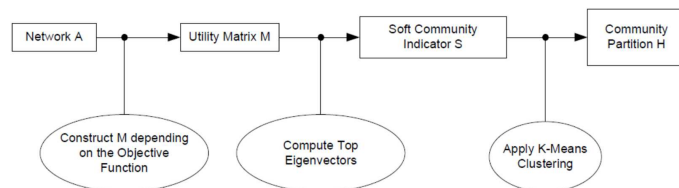
**Low** computational complexity (not  $n^2$  but  $n \log n$ ).  
118M nodes/1B links in 152mn

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## Community detection algorithms

### A Unified View for Community Partition

- Latent space models, block models, spectral clustering, and modularity maximization can be unified as



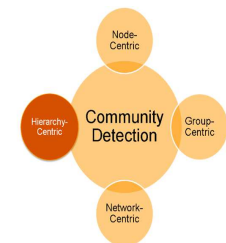
$$\text{Utility Matrix } M = \begin{cases} \text{modified proximity matrix } \tilde{P} & \text{if latent space models} \\ \text{adjacency matrix } A & \text{if block models} \\ \text{graph Laplacian } \tilde{L} & \text{if spectral clustering} \\ \text{modularity maximization } B & \text{if modularity maximization} \end{cases}$$

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## Community detection algorithms

Build a hierarchical structure of communities based on network topology

- Facilitate the analysis at different resolutions
- Representative Approaches:
  - Agglomerative Hierarchical Clustering
  - **Divisive Hierarchical Clustering**



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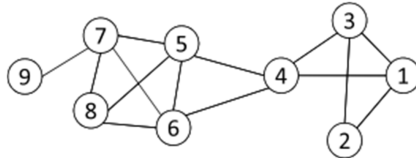
## Community detection algorithms

### Divisive clustering on Edge-Betweenness

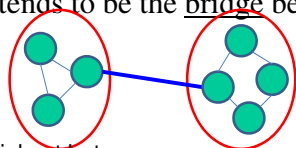
#### Hypothesis

The strength of a tie can be measured by **edge betweenness**

- **Edge betweenness**: the number of shortest paths that pass along with the edge



- The edge with higher betweenness tends to be the bridge between two communities.

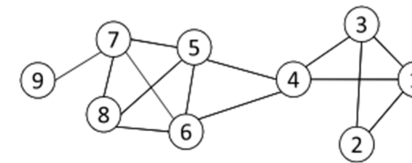


Idea: progressively removing edges with the highest betweenness

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## Community detection algorithms

### Divisive clustering based on edge betweenness

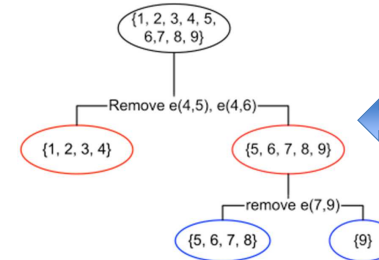


Initial betweenness values  
e(4,5) and e(4,6) have the highest betweenness



After remove e(4,5), the numerator of betweenness of e(4, 6) becomes 20, which is the highest;

After remove e(4,6), the edge e(7,9) has the highest betweenness value 4, and should be removed.



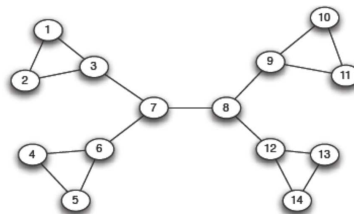
Idea: progressively removing edges with the highest betweenness

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## Community detection algorithms

### Exercise - Divisive clustering based on edge betweenness

Given the following graph, propose a clustering of the vertices based on edge betweenness



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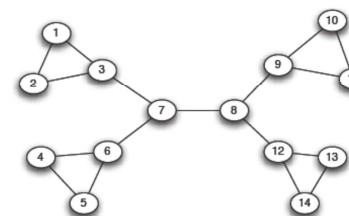
## Community detection algorithms

### Divisive clustering based on edge betweenness

Idea: progressively removing edges with the highest betweenness

Initial betweenness value

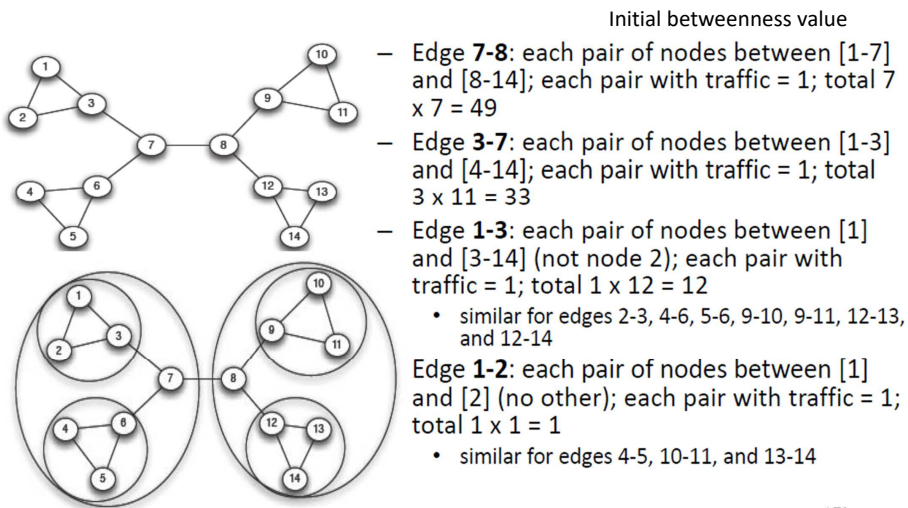
- Edge 7-8: each pair of nodes between [1-7] and [8-14]; each pair with traffic = 1; total  $7 \times 7 = 49$
- Edge 3-7: each pair of nodes between [1-3] and [4-14]; each pair with traffic = 1; total  $3 \times 11 = 33$
- Edge 1-3: each pair of nodes between [1] and [3-14] (not node 2); each pair with traffic = 1; total  $1 \times 12 = 12$ 
  - similar for edges 2-3, 4-6, 5-6, 9-10, 9-11, 12-13, and 12-14
- Edge 1-2: each pair of nodes between [1] and [2] (no other); each pair with traffic = 1; total  $1 \times 1 = 1$ 
  - similar for edges 4-5, 10-11, and 13-14



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## Community detection algorithms

### Divisive clustering based on edge betweenness



Idea: progressively removing edges with the highest betweenness

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## Community detection algorithms

### Agglomerative clustering

- Initialize each node as a community
- Choose two communities satisfying certain **criteria** and merge them into larger ones
  - Maximum Modularity Increase
  - Maximum Node Similarity

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## Community detection algorithms

### Evaluating Community Detection - 4 cases

- With formal definition of the notion of community
- With ground truth information
- Without ground truth
- With contextual information

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## Community detection algorithms

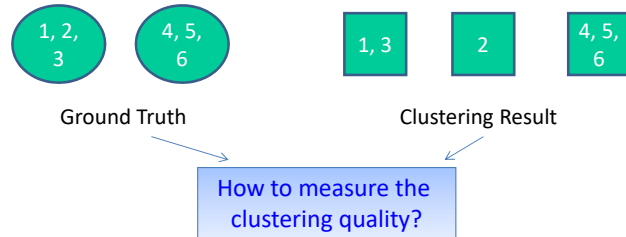
### Evaluating Community Detection

- **For community with formal definition**
  - E.g., Cliques, k-cliques, k-clubs, quasi-cliques
  - Verify whether extracted communities satisfy the definition
- **For networks with ground truth information**
  - Normalized mutual information (or ARI)
  - Accuracy of pairwise community memberships

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# Community detection algorithms

## Measuring a Clustering Result



- The number of communities after grouping can be different from the ground truth
- No clear community correspondence between clustering result and the ground truth
- Normalized Mutual Information can be used

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# Community detection algorithms

## Normalized Mutual Information

- Entropy**: the information contained in a distribution

$$H(X) = \sum_{x \in X} p(x) \log p(x)$$

- Mutual Information**: the shared information between two distributions

$$I(X; Y) = \sum_{y \in Y} \sum_{x \in X} p(x, y) \log \left( \frac{p(x, y)}{p_1(x)p_2(y)} \right)$$

- Normalized Mutual Information** (between 0 and 1)

$$NMI(X; Y) = \frac{I(X; Y)}{\sqrt{H(X)H(Y)}} \quad \text{JMLR03, Strehl} \quad \text{or} \quad NMI(X; Y) = \frac{2I(X; Y)}{H(X) + H(Y)} \quad \text{KDD04, Dhillon}$$

- Consider a partition as a distribution (probability of one node falling into one community), we can compute the matching between the clustering result and the ground truth

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## NMI

$$H(X) = \sum_{x \in X} p(x) \log p(x)$$

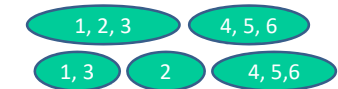
$$I(X; Y) = \sum_{y \in Y} \sum_{x \in X} p(x, y) \log \left( \frac{p(x, y)}{p_1(x)p_2(y)} \right) \rightarrow I(\pi^a, \pi^b) = \sum_h \sum_\ell \frac{n_{h,\ell}}{n} \log \left( \frac{\frac{n_{h,\ell}}{n}}{\frac{n_h^a}{n} \frac{n_\ell^b}{n}} \right)$$

$$NMI(X; Y) = \frac{I(X; Y)}{\sqrt{H(X)H(Y)}} \rightarrow NMI(\pi^a, \pi^b) = \frac{\sum_{h=1}^{k^{(a)}} \sum_{\ell=1}^{k^{(b)}} n_{h,\ell} \log \left( \frac{n \cdot n_{h,\ell}}{n_h^a \cdot n_\ell^b} \right)}{\sqrt{\left( \sum_{h=1}^{k^{(a)}} n_h^a \log \frac{n_h^a}{n} \right) \left( \sum_{\ell=1}^{k^{(b)}} n_\ell^b \log \frac{n_\ell^b}{n} \right)}}$$

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## NMI-Example

- Partition a: [1, 1, 1, 2, 2, 2]
- Partition b: [1, 2, 1, 3, 3, 3]



| $n = 6$       | $n_h^a$ | $n_\ell^b$ | $n_{h,\ell}$ | l=1 | l=2 | l=3 |
|---------------|---------|------------|--------------|-----|-----|-----|
| $k^{(a)} = 2$ | h=1     | 3          | l=1          | 2   | 1   | 0   |
| $k^{(b)} = 3$ | h=2     | 3          | l=2          | 1   | 0   | 3   |
|               |         |            | l=3          | 3   |     |     |

contingency table or confusion matrix

$$NMI(\pi^a, \pi^b) = \frac{\sum_{h=1}^{k^{(a)}} \sum_{\ell=1}^{k^{(b)}} n_{h,\ell} \log \left( \frac{n \cdot n_{h,\ell}}{n_h^a \cdot n_\ell^b} \right)}{\sqrt{\left( \sum_{h=1}^{k^{(a)}} n_h^a \log \frac{n_h^a}{n} \right) \left( \sum_{\ell=1}^{k^{(b)}} n_\ell^b \log \frac{n_\ell^b}{n} \right)}} = 0.8278$$

Reference: <http://www.cse.ust.hk/~weikep/notes/NormalizedMI.m>

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## Accuracy of Pairwise Community Memberships

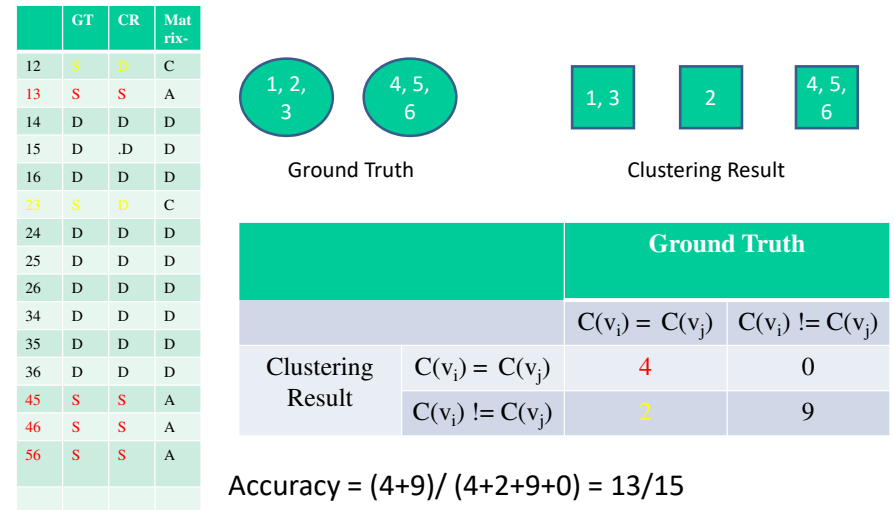
- Consider **all the possible pairs of nodes** and check whether they reside in the same community
- An **error** occurs *if*
  - Two nodes belonging to the **same** community are assigned to **different** communities after clustering
  - Two nodes belonging to **different** communities are assigned to the **same** community
- Construct a **contingency table or confusion matrix**

| Clustering Result | Ground Truth      |                      |
|-------------------|-------------------|----------------------|
|                   | $C(v_i) = C(v_j)$ | $C(v_i) \neq C(v_j)$ |
|                   | $C(v_i) = C(v_j)$ | $C(v_i) \neq C(v_j)$ |
|                   | a                 | b                    |
|                   | c                 | d                    |

$$\text{Accuracy} = \frac{a+d}{a+b+c+d} = \frac{a+d}{n(n-1)/2}$$

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## Accuracy Example



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## Evaluation using Semantics

- For networks with semantics
  - Networks come with semantic or attribute information of nodes or connections
  - Human subjects can verify whether the extracted communities are coherent
- Evaluation is **qualitative**
- It is also intuitive and helps understand a community



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## Evaluation without Ground Truth

- For networks without ground truth or semantic information
- This is the most common situation and the most difficult
- An option is to resort to **cross-validation but not suited on network with specific structure (ie preferential attachment)**.
  - Extract communities from a (training) network
  - Evaluate the quality of the community structure on a network constructed from a different date or based on a related type of interaction
- Quantitative evaluation functions
  - Modularity (M.Newman. Modularity and community structure in networks. PNAS 06.)
  - Use generative models able to learn the structure of the network and the community then compare the predicted network with the original network (Stochastic block models).

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## Community detection: challenges

- Bipartite graph

- Fuzzy community [Zhou,2015]

- Community in attributed graph

Moser et al. (2007);Zhou et al. (2009); Li et al. (2008); Cruz Gomez et al. (2011); Combe et al. (2012); Dang et Viennet (2012)

- Overlapping community

Jierni Xie, Stephen Kelley, Boleslaw K. Szymanski

Overlapping Community Detection in Networks: the State of the Art and Comparative Study  
M. G. J. Baumes, M.Magdon-Ismail, (2005) , I. F. G. Palla, I. Derenyi, T. Vicsek, Nature, 2005.

- Hierarchical community

- Yang,2013

- Dynamic graph

- Palla, Barabasi and Vicsek, Nature 2007, Hopcroft, Khan, Kulis and Selman, PNAS 2004, Jdida, Robardet and Fleury, ICDIM 2007, D. Chakrabarti, R. Kumar, and A. Tomkins, SIGKDD 2006, Y. Lin et al, Transactions on Knowledge Discovery from Data

## Community detection: challenges

- Community in attributed graph

Moser et al. (2007);Zhou et al. (2009); Li et al. (2008); Cruz Gomez et al. (2011); Combe et al. (2012); Dang et Viennet (2012)

Justified by two phenomena observed in social networks

- Homophily: tendency to stay together / have relationships with people similar to you [McPherson et al. 2001]
- Social influence: Ability to change the behavior and/or opinion of the others / neighbors [Anagnostopoulos,2008 Crandall et al. 2008]
- Consequences:
  - ▶ A node is more likely to be similar to its neighbors
  - ▶ Exploit both links and attributes can help to detect communities

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## Outline

- Introduction
- Networks representation, storage, visualization
- Network analysis : measures and metrics
- Models of network generation
- Node classification
- Community detection
- Link mining task/ Influence analysis
- Conclusion
- References
- Application with R

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## Link mining tasks

Various tasks :

- Object-Related Tasks
  - Link-based object ranking (LBR)
  - Link-based object classification (LBC)
  - Object identification (entity resolution)
- Link-Related Tasks
  - Link prediction
  - Link prediction variants
- Graph-Related Tasks
  - Subgraph discovery
  - Graph classification

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## Link mining tasks

### Link-based object ranking LBR

- Link-based object ranking (LBR)- definition
  - LBR is a core analysis task with objective: rank individuals in terms of “centrality”
  - Exploit the link structure of a graph to order or prioritize the set of objects within the graph
  - Focused on graphs with single object type and single link type
- Example: Web information analysis
  - PageRank and Hits are typical LBR approaches
- Variants: Rank objects relative to one or more relevant objects in the graph vs. ranks object over time in dynamic graphs

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## Link mining tasks

### Link-based object classification (LBC)

- LBC definition
  - Predicting the category of an object based on its attributes, its links and the attributes of linked objects
- Hypothesis
  - Labels of related objects tend to be correlated
  - Collective classification: Explore such correlations and jointly infer the categorical values associated with the objects in the graph
- Examples
  - Web:** Predict the category of a web page, based on words that occur on the page, links between pages, anchor text, html tags, etc.
  - Citation:** Predict the topic of a paper, based on word occurrence, citations, co-citations
  - Authors:** Predict the research field based on paper content, co-authors

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## Link mining tasks

### Object identification (entity resolution)

- Definition
  - Predicting when two objects are the same, based on their attributes and their links
  - Also known as: deduplication, reference reconciliation, co-reference resolution, object consolidation
- Applications
  - Web:** predict when two sites are mirrors of each other
  - People :** predicting when two names are referring to the person
    - Jean Monnet University,
    - Université Jean Monnet,
    - Université de Saint-Etienne

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## Link mining tasks

### Object identification (entity resolution)

- Earlier viewed as pair-wise resolution problem: resolved based on the similarity of their attributes (soft tf idf, monge)
- Importance at considering links
  - Coauthor links in bib data, hierarchical links between spatial references, co-occurrence links between name references in documents
- Use of links in resolution
  - Collective entity resolution: one resolution decision affects another if they are linked
    - Propagating evidence over links in a depen. graph
  - Probabilistic models interact with different entity recognition decisions

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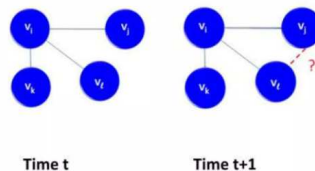
## Link mining tasks

**Aim of link prediction:** predict the presence or absence of a relationship between pairs of nodes in a social network.

**Input:** A graph  $G(V, E)$  and a list of unlabeled pairs of nodes  $(u, v)$  in  $V \times V$

**Output:** a binary label taking value 1 if the nodes are connected, 0 otherwise.

**Interest:** Prediction of the appearance of a link in a dynamic network or in bipartite graph for recommendation.



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## Link mining tasks

Basic solution:

- Assign a score to every possible link
- Sort the links by descending orders of the score
- Predict the top-k links or the links with score above a threshold

How to define the score?

- Common neighbors (Newman2001):

$$\text{score}(u, v) = |\Gamma(u) \cap \Gamma(v)|$$

- Based on a graph-based distance between the nodes (geodesic distance, Jaccard, Adamic, Katz 1953, Simrank [Jeh 2002], etc.)
- Based on a vector-based distance after embedding the nodes

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## Link mining tasks

Formulate the link prediction problem as a supervised classification problem [Hassan, 2006]:

Given a training and a test sets composed of pairs of links described by **predictive features** and the label 1 if they are connected, 0 otherwise

- Learn a classification model on the training set
- Evaluate the model on a test set or by cross-validation
- Apply the model to predict the presence/absence of a link for the unlabeled pairs of nodes

Predictive features:

- Characteristics of the nodes belonging to the pairs (degree, clustering coefficient, similarity in the graph, etc) or edges
- Embedding representations
- etc.

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## Link mining tasks

### Link prediction variants

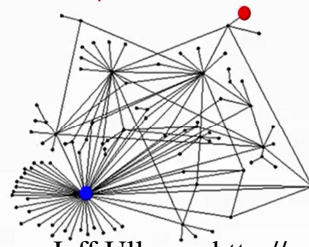
- Predict the weighted link between a pair of nodes: 0 /  $w(u, v)$
- Predicting the number of links to an object
  - **Web:** predict the authority of a page based on the number of in-link; identifying hubs based on the number of out-links (Page rank, Hubs and authorities: HITS)
  - **Citation:** predicting the impact of a paper based on the number of citations
  - **Epidemics:** predicting the number of people that will be infected based on the infectiousness of a disease
- Predicting the number of objects reached along a path from an object
  - **Web:** predicting number of pages retrieved by crawling a site
  - **Citation:** predicting the number of citations of a particular author in a specific journal

## Link mining tasks: Page rank

- **How to organize the Web?**
- **First try: Human curated Web directories**
  - Yahoo, DMOZ, LookSmart
- **All web pages are not equally "important"**  
[www.joe-schmoe.com](http://www.joe-schmoe.com) vs. [www.stanford.edu](http://www.stanford.edu)



- There is large diversity in the web-graph node connectivity.  
**Let's rank the pages by the link structure!**



@Mining of Massive Datasets

Jure Leskovec, Anand Rajaraman, Jeff Ullman <http://www.mmids.org/>

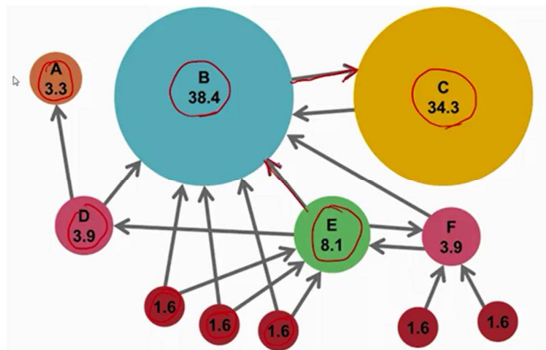
## Page rank: underlying idea

- **Idea: Links as votes**
  - Page is more important if it has more links
    - In-coming links? Out-going links?
- **Think of in-links as votes:**
  - [www.stanford.edu](http://www.stanford.edu) has 23,400 in-links
  - [www.joe-schmoe.com](http://www.joe-schmoe.com) has 1 in-link
- **Are all in-links are equal?**
  - Links from important pages count more
  - Recursive question!

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## Page rank: underlying idea

- A vote from an important page is worth more
- A page is more important if it is pointed by other important pages



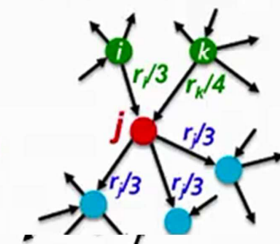
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## Page rank: simple recursive formulation

- Each link's vote is proportional to the **importance** of its source page
- If page  $j$  with importance  $r_j$  has  $n$  out-link each link gets  $r_j/n$  votes
- Page  $j$ 's own importance is the sum of the votes on its in-links

Recursive formulation:

$$r_j = r_i/3 + r_k/4$$



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# Page rank: simple recursive formulation

## Recursive formulation:

- Define a "rank"  $r_j$  for page  $j$

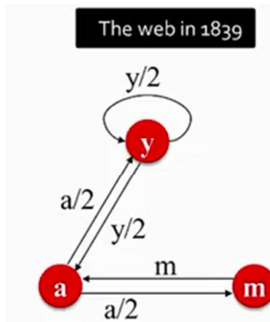
$$r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$$

$d_i$  ... out-degree of node  $i$

- 3 equations, 3 unknowns, no constants**

→ No unique solution

- All solutions equivalent modulo the scale factor



Flow equations:  
 $r_y = r_y/2 + r_a/2$   
 $r_a = r_y/2 + r_m$   
 $r_m = r_a/2$

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# Page rank: simple recursive formulation

## Recursive formulation:

- Define a "rank"  $r_j$  for page  $j$

$$r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$$

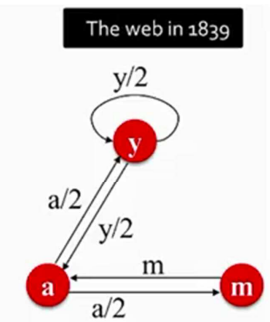
$d_i$  ... out-degree of node  $i$

- Additional constraint forces uniqueness:**

$$r_y + r_a + r_m = 1$$

→ **Solution:**  $r_y = \frac{2}{5}$ ,  $r_a = \frac{2}{5}$ ,  $r_m = \frac{1}{5}$

- Gaussian elimination method works for small examples, but we need a better method for large web-size graphs**
- We need a new formulation!**



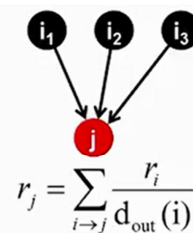
"Flow" equations:  
 $r_y = r_y/2 + r_a/2$   
 $r_a = r_y/2 + r_m$   
 $r_m = r_a/2$

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# Page rank: power iteration method based on Markov process simulating a random surfer

- Imagine a random web surfer:**

- At any time  $t$ , surfer is on some page  $i$
- At time  $t + 1$ , the surfer follows an out-link from  $i$  uniformly at random
- Ends up on some page  $j$  linked from  $i$
- Process repeats indefinitely

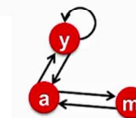


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# Page rank: power iteration method based on Markov process

- Transition matrix M**

- Let page  $i$  has  $d_i$  out-links
- If  $i \rightarrow j$ , then  $M_{ji} = \frac{1}{d_i}$  else  $M_{ji} = 0$ 
  - $M$  is a **column stochastic matrix**
    - Columns sum to 1



|   | y   | a   | m |
|---|-----|-----|---|
| y | 1/2 | 1/2 | 0 |
| a | 1/2 | 0   | 1 |
| m | 0   | 1/2 | 0 |

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## Page rank: power iteration method based on Markov process

- Transition matrix  $M$ : describes what's happen to a random surfer after one step

### Transition matrix $M$

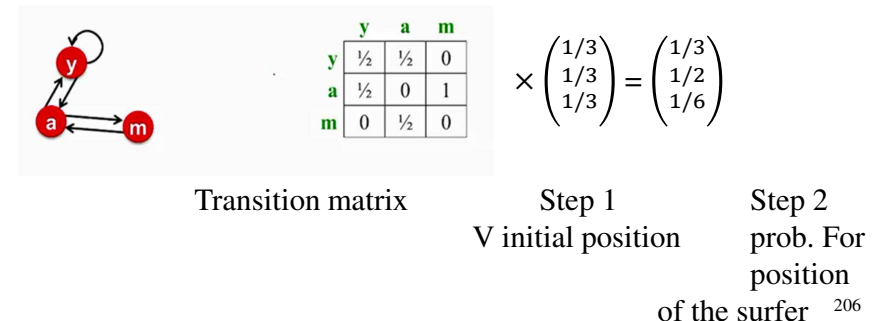
- Let page  $i$  has  $d_i$  out-links
- If  $i \rightarrow j$ , then  $M_{ji} = \frac{1}{d_i}$  else  $M_{ji} = 0$ 
  - $M$  is a **column stochastic matrix**
    - Columns sum to 1

- The probability distribution for the location of the surfer is a vector  $v$  column having  $1/N$  for each component if we suppose that the surfer starts at any node with equal probability.
- At step 2, the distribution is given by:  $p = Mv$

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## Page rank: power iteration method based on Markov process

- Equal probability to start at every node
- The probability distribution at next step is given by  $p(t+1) = M p(t)$



## Page rank: power iteration method / Markov process

### Let:

- $p(t)$  ... vector whose  $i^{\text{th}}$  coordinate is the prob. that the surfer is at page  $i$  at time  $t$
- So,  $p(t)$  is a probability distribution over pages

- Transition matrix  $M$ : describes what's happen to a random surfer after one step

### Transition matrix $M$

- Let page  $i$  has  $d_i$  out-links
- If  $i \rightarrow j$ , then  $M_{ji} = \frac{1}{d_i}$  else  $M_{ji} = 0$ 
  - $M$  is a **column stochastic matrix**
    - Columns sum to 1

**Markov Process:** Under conditions on  $M$ , the distribution of the surfer converges to a limiting distribution defined by:  $p = Mp$

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## Page rank: power iteration method based on Markov process

### Power Iteration Method

- Given a web graph with  $N$  nodes, where the nodes are pages and edges are hyperlinks**
- Power iteration:** a simple iterative scheme

- Suppose there are  $N$  web pages
- Initialize:  $r^{(0)} = [1/N, \dots, 1/N]^T$
- Iterate:  $r^{(t+1)} = M \cdot r^{(t)}$
- Stop when  $|r^{(t+1)} - r^{(t)}|_1 < \epsilon$

$$r_j^{(t+1)} = \sum_{i \rightarrow j} \frac{r_i^{(t)}}{d_i}$$

$d_i$  .... out-degree of node  $i$

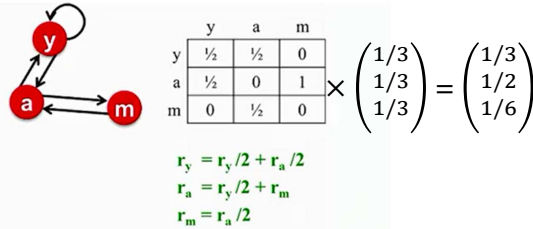
$|x|_1 = \sum_{1 \leq i \leq N} |x_i|$  is the  $L_1$  norm  
Can use any other vector norm, e.g., Euclidean

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## Page rank: power iteration method based on Markov process

### Power Iteration:

- Set  $r_j = 1/N$
- 1:  $r'_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$
- 2:  $r = r'$
- If not converged: goto 1



### Example:

$$\begin{bmatrix} r_y \\ r_a \\ r_m \end{bmatrix} = \begin{bmatrix} 1/3 & 1/3 & 5/12 & 9/24 & & 6/15 \\ 1/3 & 3/6 & 1/3 & 11/24 & \dots & 6/15 \\ 1/3 & 1/6 & 3/12 & 1/6 & & 3/15 \end{bmatrix}$$

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## Page rank: power iteration method based on Markov process

- A central result from the theory of random walks (a.k.a. Markov processes):

For graphs that satisfy **certain conditions**, the **stationary distribution is unique** and eventually will be reached no matter what the initial probability distribution at time  $t = 0$

$$r_j^{(t+1)} = \sum_{i \rightarrow j} \frac{r_i^{(t)}}{d_i} \quad \text{or equivalently} \quad r = Mr$$

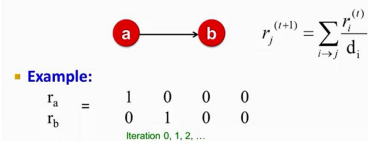
- Does this converge?
- Does it converge to what we want?

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## Page rank: 2 issues

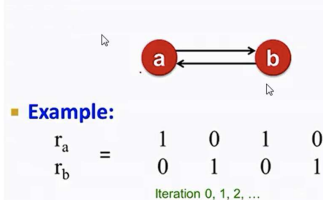
- Convergence conditions and issues on the web :
  - There is no node that has no arc out : Dead end issue

### The "Dead end" problem:



- Graph is strongly connected: Spider trap issue

### The "Spider trap" problem:



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## Page rank: 2 issues

### Problem: Dead Ends

### Power Iteration:

- Set  $r_j = 1$
- $r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$
- And iterate

### Example:

$$\begin{bmatrix} r_y \\ r_a \\ r_m \end{bmatrix} = \begin{bmatrix} 1/3 & 2/6 & 3/12 & 5/24 & & 0 \\ 1/3 & 1/6 & 2/12 & 3/24 & \dots & 0 \\ 1/3 & 1/6 & 1/12 & 2/24 & & 0 \end{bmatrix}$$

Iteration 0, 1, 2, ...

- Solutions :
- Avoiding dead ends
  - Teleporting

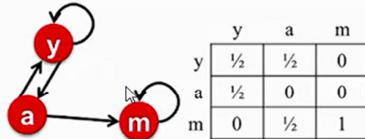
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## Page rank: 2 issues

### Problem: Spider Traps

#### Power Iteration:

- Set  $r_j = 1$
- $r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$
- And iterate



$$\begin{aligned} r_y &= r_y/2 + r_a/2 \\ r_a &= r_y/2 \\ r_m &= r_a/2 + r_m \end{aligned}$$

#### Example:

$$\begin{bmatrix} r_y \\ r_a \\ r_m \end{bmatrix} = \begin{bmatrix} 1/3 & 2/6 & 3/12 & 5/24 & & 0 \\ 1/3 & 1/6 & 2/12 & 3/24 & \dots & 0 \\ 1/3 & 3/6 & 7/12 & 16/24 & & 1 \end{bmatrix}$$

Iteration 0, 1, 2, ...

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## Page rank with teleporting

### Solution: Random Teleports

- The Google solution for spider traps: **At each time step, the random surfer has two options**
  - With prob.  $\beta$ , follow a link at random
  - With prob.  $1-\beta$ , jump to some random page
  - Common values for  $\beta$  are in the range 0.8 to 0.9
- Surfer will teleport out of spider trap within a few time steps**

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## Page rank with teleporting

### Solution: Random Jumps

- Google's solution that does it all:**
  - Makes **M** stochastic, aperiodic, irreducible
- At each step, random surfer has two options:**
  - With probability  $\beta$ , follow a link at random
  - With probability  $1-\beta$ , jump to some random page
- PageRank equation** [Brin-Page, 98]

$$r_j = \sum_{i \rightarrow j} \beta \frac{r_i}{d_i} + (1 - \beta) \frac{1}{n}$$

$d_i$  ... out-degree of node i

The above formulation assumes that **M** has no dead ends. We can either preprocess matrix **M** (bad!) or explicitly follow random teleport links with probability 1.0 from dead-ends.

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## Page rank with teleporting

- PageRank equation** [Brin-Page, 98]

$$r_j = \sum_{i \rightarrow j} \beta \frac{r_i}{d_i} + (1 - \beta) \frac{1}{n}$$

- The Google Matrix A:**

$$A = \beta M + (1 - \beta) \frac{1}{n} \mathbf{e} \cdot \mathbf{e}^T$$

$\mathbf{e}$  ... vector of all 1s

- A is stochastic, aperiodic and irreducible, so**

$$\mathbf{r}^{(t+1)} = \mathbf{A} \cdot \mathbf{r}^{(t)}$$

- What is  $\beta$ ?**

- In practice  $\beta = 0.8, 0.9$  (make 5 steps and jump)

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## Influence analysis

**Basic assumption:** word-of-mouth effect, thanks to which actions, opinions, buying behaviors, innovations and so on, propagate in a social network.

**Homophilly versus social influence** Crandall et al Anagnostopoulos et al. (KDD'08)

### Homophily

tendency to stay together with people similar to you

*"Birds of a feather flock together"*

E.g., I'm overweight → I date overweight girls

### Social influence

a force that person A (i.e., the influencer) exerts on person B to introduce a change of the behavior and/or opinion of B

Influence is a causal process

E.g., my girlfriend gains weight → I gain weight too

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## Influence analysis

- Influence Maximization problem : find a seed-set of influential people such that by targeting them we maximize the spread of propagations

- Hot topic in Data Mining research since 20 years:

- Domingos and Richardson *"Mining the network value of customers"* (KDD'01)
- Domingos and Richardson *"Mining knowledge-sharing sites for viral marketing"* (KDD'02)
- Kempe et al. *"Maximizing the spread of influence through a social network"* (KDD'03)

Some of the many startups involved in social influence

Klout (<http://klout.com>)

Measure of overall influence online (mostly twitter, now FB and linkedin)

Score = function of true reach, amplification probability and network influence

Claims score to be highly correlated to clicks, comments and retweets

Peer Index (<http://www.peerindex.net>)

Identifies/scores authorities on the social web by topic

SocialMatica (<http://www.socialmatica.com>)

Ranks 32M people by vertical/topic, claims to take into account quality of authored content

Influencer50 (<http://www.influencer50.com>)

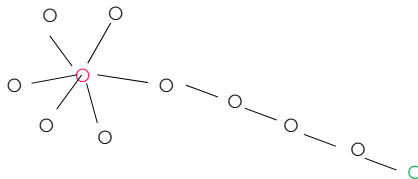
Clients: IBM, Microsoft, SAP, Oracle and a long list of tech companies

Svnetwork, Bluecalypso, CrowdBooster, Sproutsocial, TwentyFeet, EmpireAvenue, Twitaholic (there's more ...)

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## Influence analysis

- Influence Maximization problem : find a seed-set of influential people such that by targeting them we maximize the spread of propagations



## Influence Maximization Problem

following Kempe et al. (KDD'03) *"Maximizing the spread of influence through a social network"*

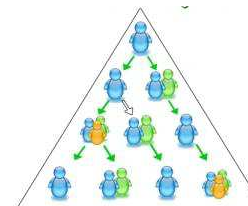
Given a **propagation model**  $M$ , define **influence** of node set  $S$ ,  
 $\sigma_M(S)$  = **expected** size of propagation, if  $S$  is the initial set of active nodes

**Problem:** Given social network  $G$  with arcs probabilities/weights, budget  $k$ , find  $k$ -node set  $S$  that maximizes  $\sigma_M(S)$

Two major **propagation models** considered:

**independent cascade (IC)** model

**linear threshold (LT)** model



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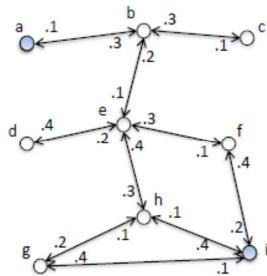
## Influence analysis

- Independent cascade (IC) model

Every arc  $(u,v)$  has associated the probability  $p(u,v)$  of  $u$  influencing  $v$

Time proceeds in discrete steps

At time  $t$ , nodes that became active at  $t-1$  try to activate their inactive neighbors, and succeed according to  $p(u,v)$



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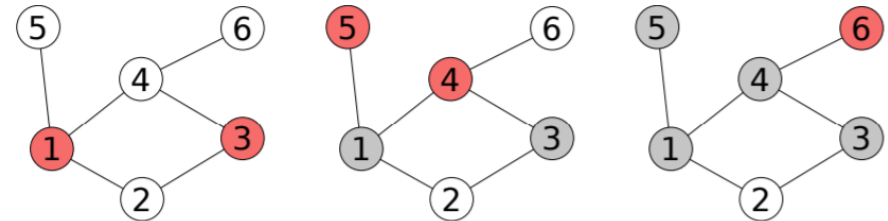
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## Influence analysis

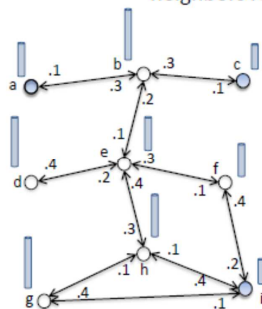
- Linear threshold model

Every arc  $(u,v)$  has associated a weight  $b(u,v)$  such that the sum of incoming weights in each node is  $\leq 1$

Time proceeds in discrete steps

Each node  $v$  picks a random threshold  $\vartheta_v \sim U[0,1]$

A node  $v$  becomes active when the sum of incoming weights from active neighbors reaches  $\vartheta_v$



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## Influence analysis

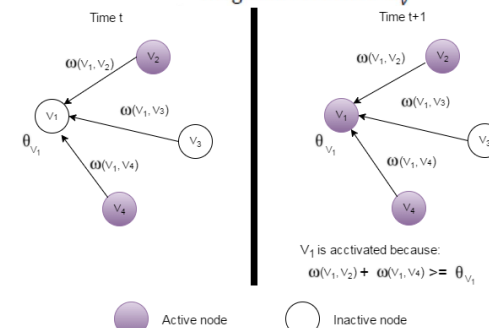
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A node  $v$  becomes active when the sum of incoming weights from active neighbors reaches  $\vartheta_v$



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Bad news: **NP-hard** optimization problem for both IC and LT models

Good news: we can use **Greedy algorithm**

---

**Algorithm 1** Greedy

---

**Input:**  $G, k, \sigma_m$

**Output:** seed set  $S$

```
1:  $S \leftarrow \emptyset$ 
2: while  $|S| < k$  do
3:   select  $u = \arg \max_{w \in V \setminus S} (\sigma_m(S \cup \{w\}) - \sigma_m(S))$ 
4:    $S \leftarrow S \cup \{u\}$ 
```

---

$\sigma_M(S)$  is **monotone** and **submodular**

**Theorem\*:** The resulting set  $S$  activates at least  $(1 - 1/e) > 63\%$  of the number of nodes that any size- $k$  set could activate

Bad news: computing  $\sigma_M(S)$  is **#P-hard** under both IC and LT models  
step 3 of the **Greedy Algorithm** above can only be approximated by MC simulations

\*Nemhauser et al. "An analysis of approximations for maximizing submodular set functions - I" (1978)

## Influence analysis

- Limits of linear / cascade model
- Do not take into account
  - The characteristics of the nodes
  - the dynamic of the network
- Extensions : Lagnier,13, Saito,09,10,11

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- **Conclusion - perspectives**
- References
- Application with R

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## Conclusion

- Many domains, many approaches, many softwares (sociology, computer science, mathematics and physics).
- Complementarity approches
- Many applications
- Need of Social Network Mining Tools (Gartner report 2008)

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# Conclusion



Providing software solutions since 1976

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NEWS EVENTS CONSULTING CAREERS RESOURCES

Home Products & Solutions Customer Success Partners Company Support & Training

PRODUCTS & SOLUTIONS / SOCIAL NETWORK ANALYSIS

## Products and Solutions

- Industries
- Small to Midsize Business
- Nonprofit Organizations
- Analytics
- Business Analytics
- Business Intelligence
- Customer Intelligence
- Data Management
- Financial Intelligence
- Foundation Tools
- Fraud Prevention & Detection
- Governance, Risk & Compliance
- High-Performance Computing
- Human Capital Intelligence
- IT Management
- OnDemand Solutions

## SAS® Social Network Analysis

An integral component of the SAS Fraud Framework

Overview Benefits Features Demos & Screenshots

Fraud is a growing problem for banks and insurance companies, in large part because it is so hard to detect. That's because fraudsters act much like legitimate customers, and they employ increasingly sophisticated fraud techniques that enable them to fly well below the radar.

SAS Social Network Analysis helps institutions detect and **prevent fraud** by going beyond individual and account views to analyze all related activities and relationships at a network dimension.

## Benefits

## Product Demo

This executive-level demonstration shows SAS Social Network Analysis being used as part of the SAS Fraud Framework to identify potential fraud in a bank.



## Questions?

Phone  
Contact Form

## Special Report

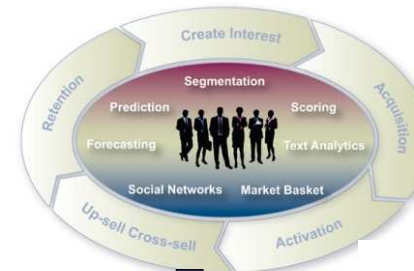
**Crimes of financial passion**  
High-tech practices to beat enterprise fraudsters at their own game

Register Now

SAS® Social Network Analysis

Overview

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## Social Network Analysis

Social networks are not just about the Internet. Your customers are a social network too. KSN identifies the otherwise hidden links - call records or bank transfers for instance - between friends, families, co-workers and other communities and extracts significant social metrics, pinpointing who are the best connected and who plays the most important role in any group. In this way it reveals valuable new customer intelligence that - when added to existing customer information - can strengthen significantly user organizations' customer acquisition, retention, cross-sell and up-sell campaigns.

More technical details about the module included: [KSN](#).

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# Conclusion

## Social network analysis softwares

### List available on Wikipedia

[http://en.wikipedia.org/wiki/Social\\_network\\_analysis\\_software](http://en.wikipedia.org/wiki/Social_network_analysis_software)

### Benchmark studies

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- Huisman et al., *Software for social network analysis* (2005)
  - A statistical point of view
  - StOCNETsoftware
- K. Xu et al., *Second International Conference on Communication Software and Networks* (2010)
  - Scalability (time and memory) point of view
  - Different types of nodes

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# Conclusion

## Social network analysis softwares

| Software                                                     | Paik [4]                                                           | Gephi [3]                                | NetworkX [21]                                                      | igraph [10]                                                |
|--------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------------|--------------------------------------------------------------------|------------------------------------------------------------|
| Version                                                      | 1.26                                                               | 0.7 alpha                                | 0.6                                                                | 0.5.3                                                      |
| Website (http://)                                            | <a href="http://paik.infim.si/doku.php">paik.infim.si/doku.php</a> | <a href="http://gephi.org">gephi.org</a> | <a href="http://igraph.sourceforge.net">igraph.sourceforge.net</a> | <a href="http://networkx.lanl.gov/">networkx.lanl.gov/</a> |
| Type                                                         | Stand-alone software                                               | Stand-alone software                     | Library                                                            | Library                                                    |
| Platform                                                     | Windows                                                            | Java                                     | Python                                                             | R / Python / C libraries                                   |
| License                                                      | Free for non-commercial use                                        | GNU GPL                                  | BSD License                                                        | GNU GPL                                                    |
| Computing time                                               | Fast (C)                                                           | Medium (Java)                            | Fast (C, Python)                                                   | Fast (C)                                                   |
| Tractable number of nodes                                    | 500,000 nodes                                                      | 150,000 nodes                            | 1,000,000 nodes                                                    | > 1.9 million relations (without attributes)               |
| Time to load 10 <sup>5</sup> nodes and 10 <sup>6</sup> edges | 24 seconds                                                         | 40 seconds                               | 137 seconds                                                        | 11 seconds                                                 |
| Ergonomics                                                   |                                                                    |                                          |                                                                    |                                                            |
| Ease of handling                                             | Good                                                               | Good                                     | Good (but Python handling needed)                                  | Average (R handling recommended)                           |
| File formats                                                 |                                                                    |                                          |                                                                    |                                                            |
| GML                                                          | X                                                                  | ✓                                        | ✓                                                                  | ✓                                                          |
| Paik (.net)                                                  | ✓                                                                  | Import only                              | ✓                                                                  | ✓                                                          |
| GraphML                                                      | Export only                                                        | ✓                                        | ✓                                                                  | ✓                                                          |
| DL                                                           | ✓                                                                  | ✓                                        | X                                                                  | X                                                          |
| Graph types                                                  |                                                                    |                                          |                                                                    |                                                            |
| Two-mode graphs                                              | ✓                                                                  | X                                        | ✓                                                                  | ✓                                                          |
| Multi-relational graphs                                      | ✓                                                                  | X                                        | X                                                                  | X                                                          |
| Temporality                                                  | ✓                                                                  | X                                        | ✓                                                                  | X                                                          |
| Visualization layouts                                        |                                                                    |                                          |                                                                    |                                                            |
| Fruchterman Reingold                                         | ✓                                                                  | ✓                                        | X                                                                  | ✓                                                          |
| Kamada Kawai                                                 | ✓                                                                  | ✓                                        | X                                                                  | ✓                                                          |
| Other spring layouts                                         | X                                                                  | ✓                                        | ✓                                                                  | ✓                                                          |
| Indicators                                                   |                                                                    |                                          |                                                                    |                                                            |
| Degree, betweenness and closeness centrality                 | ✓                                                                  | ✓                                        | ✓                                                                  | ✓                                                          |
| Dyad census                                                  | X                                                                  | X                                        | X                                                                  | ✓                                                          |
| Triad census                                                 | ✓                                                                  | X                                        | X                                                                  | ✓                                                          |
| HITS, PageRank                                               | X                                                                  | ✓                                        | ✓                                                                  | ✓                                                          |
| Clustering algorithms                                        |                                                                    |                                          |                                                                    |                                                            |
| Edge betweenness                                             | X                                                                  | X                                        | X                                                                  | ✓                                                          |
| Walktrap                                                     | X                                                                  | X                                        | X                                                                  | ✓                                                          |
| Springlass                                                   | X                                                                  | X                                        | X                                                                  | ✓                                                          |
| Dendrogram display                                           | ✓                                                                  | ✓                                        | X                                                                  | ✓                                                          |

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## Conclusion

### Social network analysis softwares

| Criteria             | Pajek [4] | Gephi [3] | NetworkX [21] | igraph [10] |
|----------------------|-----------|-----------|---------------|-------------|
| c1 License           | –         | +         | ++            | +           |
| c2 Capacity          | +         | –         | ++            | ++          |
| c3 Indicators        | +         | ++        | ++            | ++          |
| c4 File formats      | +         | ++        | +             | +           |
| c5 Graph types       | ++        | –         | +             | –           |
| c6 Visualization     | +         | ++        | –             | ++          |
| c7 Clustering        | +         | +         | --            | ++          |
| c8 Custom attributes | +         | +         | ++            | --          |

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## Conclusion

- Large size network
  - NOSQL graph database (Neo4J, Cassandra)
  - Algorithmic Problems
    - Community detection:
      - » Multi-level graph partitioning
      - » Local graph clustering
- Complex networks
  - Different types of nodes
  - Different types of links

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## Conclusion

### Adding Content based analysis

- Links and nodes attributes
- Multimedia information :Text, image, video, etc
  - Transfer learning
- Provided by sensor (GPS)
- > mining heterogeneous information network : much more to explore
- Dynamic and Spatial analysis
  - Network evolution (characteristics, etc)
  - Node/Link/attribute prediction (Kleinberg,2003)
  - Community evolution
- Privacy / Trust / Fairness in social networks

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## Conclusion

### Future research:

- Discovering ontology and structure in information networks
- Discovering and mining hidden information networks
- Mining information networks formed by structured data linking with unstructured data (text, multimedia and Web)
- Mining cyber-physical networks (networks formed by dynamic sensors, image/video cameras, with information networks)
- Enhancing the power of knowledge discovery by transforming massive unstructured data: Incremental information extraction, role discovery, ... ⇒ multi-dimensional structured info-net
- Mining noisy, uncertain, un-trustable massive datasets by information network analysis approach
- Turning Wikipedia and/or Web into structured or semi-structured databases by heterogeneous information network analysis

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# Conclusion

## Social network analysis

## Social network mining

## Information network mining (Aggarwal,2011)

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# Outline

- Introduction
- Networks representation, storage, visualization
- Network analysis : measures and metrics
- Models of network generation
- Node classification
- Community detection
- Influence/ link prediction
- Conclusion - perspectives
- [References](#)

- Application with R

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